

INORGANIC CHEMISTRY LABORATORY MANUAL - VERSION 1.0



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Inorganic Chemistry Laboratory Manual

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Detailed Licensing

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Introduction

[1.1: Welcome to the Inorganic Chemistry Laboratory](#)

[1.2: The Laboratory Notebook](#)

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1.1: Welcome to the Inorganic Chemistry Laboratory

Welcome to the Inorganic Chemistry Laboratory. This laboratory course provides you with the opportunity to integrate the knowledge and skills you have developed throughout the chemistry curriculum. You will apply concepts from general chemistry, organic chemistry, analytical chemistry, and physical chemistry to understand the synthesis, characterization, and properties of inorganic compounds.

The primary goals of this course are to provide experience with:

- **Modern techniques for the synthesis of inorganic compounds**, which may include microscale methods, vacuum-line techniques, high-pressure reactions, and controlled-atmosphere procedures.
- **Methods for characterizing inorganic compounds**, which may include electron paramagnetic resonance (EPR), infrared (IR) spectroscopy, nuclear magnetic resonance (NMR) spectroscopy, UV-Visible spectroscopy, magnetic susceptibility measurements, electrochemistry, and powder X-ray diffraction.
- **The relationship between structure and properties**, emphasizing the fundamental principles of inorganic chemistry that govern the behavior of the compounds prepared in the laboratory.
- **Maintaining a complete and accurate laboratory notebook** that serves as a permanent record of your experimental work.
- **Writing clear, well-organized laboratory reports** using the conventions of scientific communication.
- **Locating and interpreting information from the chemical literature** and other reliable scientific resources.
- **Practicing safe laboratory techniques**, maintaining a constant awareness of chemical hazards, and fostering a culture of safety throughout every experiment.

The experiments in this manual have been selected to expose you to a broad range of synthetic methods, analytical techniques, and chemical concepts representative of inorganic chemistry. Each experiment is designed to reinforce the connections between concepts learned in previous and current lecture courses while introducing new methods and instrumentation that expand your experimental skills.

Careful preparation, thoughtful experimentation, accurate record keeping, and critical analysis of your results will be essential to your success in this course. By the end of the semester, you will have strengthened your understanding of inorganic chemistry while developing the technical skills, scientific reasoning, and communication abilities expected of a professional chemist.

Welcome to the laboratory. We hope you find the experience engaging, challenging, and fun.

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1.2: The Laboratory Notebook

Laboratory Notebook

The laboratory notebook is your permanent record of your experimental work. It must provide a complete and honest account of the procedures performed, data collected, and observations made for each experiment. Your goal should be to maintain a notebook such that someone with at least your level of experience could repeat the experiment in its entirety solely using what you have recorded in your notebook.

Students must use a bound laboratory notebook with numbered pages. Students must bring their laboratory notebook to every laboratory session. All data and observations must be recorded immediately at the time they are made directly in the laboratory notebook and not on scrap paper or elsewhere. The laboratory notebook may be inspected or collected for grading at any time during the laboratory period.

Leave the first two (or more) pages of the notebook blank for a table of contents, or, if your notebook has a pre-printed table of contents, you may use that. If the notebook pages are not already numbered, number each page sequentially. All measurements, observations, and experimental conditions must be recorded directly in the notebook using black or dark blue ink. Erasures are not acceptable. If an entry must be corrected or discarded, draw a single line through the original entry so that it remains legible. Begin recording each experiment on a new page. Record the date of each new entry.

Preparation Before Lab

Before entering the laboratory, carefully study the experiment for that day. Write the experimental procedure, reagent quantities, and experimental setup in your notebook beforehand. Prepare any equations, constants, and data tables needed for the experiment so that data can be recorded efficiently during the laboratory period.

Recording Data

Record observations and measurements clearly and sequentially so that the experiment can be easily followed. When appropriate, record both elapsed time and real time. All numerical values must be recorded with the appropriate units. During the experiment, check calculations and measurements whenever possible to ensure the consistency of your data and to identify potential errors early. Leave sufficient blank space in your notebook entries to allow additional observations, calculations, or corrections to be added later.

Calculations and Data Analysis

After all measurements and observations have been recorded, perform the required calculations and graphical analysis. These may be completed in the notebook or using appropriate software while referencing the notebook entries. If instrumental data such as spectra are obtained, attach a copy to the relevant page in the notebook.

Use of the Notebook

The laboratory notebook should not contain a complete copy of the final written report; however, it may be used to draft portions of the report. The notebook may also be used to record literature references, ideas, and other information that may be useful in future laboratory work.

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CHAPTER OVERVIEW

2: Safety

[2.1: General Safety Rules](#)

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2.1: General Safety Rules

Rules specific to each activity are provided in the introduction to that activity.

Safety must be the first concern of everyone working in the laboratory. The following rules must be followed without exception.

Personal Procedures & Safety

1. Safety goggles must be worn in the laboratory at all times. Contact lenses are not recommended. Failure to wear safety goggles as instructed will result in dismissal from the laboratory.
2. Shoes that completely cover the heel, toes, top, and sides of the feet are required. Clothing that covers the legs below the knees is required. Clothing should not be loose or flowing, especially the sleeves. Some synthetic fabrics are highly flammable and should not be worn. Long hair must be tied back and kept away from flames and chemicals.
3. A laboratory coat must be worn whenever performing laboratory experiments unless otherwise directed by your instructor. The laboratory coat must be properly fastened during laboratory work and removed before leaving the laboratory to prevent the spread of chemical contamination.
4. Wear appropriate chemical-resistant gloves whenever handling hazardous chemicals. Select gloves that are compatible with the chemicals being used, inspect them for damage before use, and remove them immediately if they become contaminated. Gloves must not be worn outside the laboratory or when touching common surfaces such as door handles, keyboards, or mobile phones.
5. Before using any unfamiliar chemical, review its Safety Data Sheet (SDS) and understand its hazards, required personal protective equipment, safe handling procedures, storage requirements, and appropriate emergency response measures.
6. Food, drinks, and chewing gum are prohibited in the laboratory.
7. If you have allergies, medical conditions (e.g., asthma), or other health concerns that could affect your safety in the laboratory, you are encouraged to inform your teaching assistant or instructor, as you feel appropriate. If an experiment involves a substance to which you are allergic or otherwise poses a health concern, consult your instructor before beginning the experiment. Permission to modify safety procedures or experimental procedures to accommodate a health concern may only be granted by the Faculty Laboratory Coordinator or the Laboratory Manager.
8. If you are pregnant or become pregnant during the semester, notify your instructor and consult your physician before performing laboratory experiments.
9. Horseplay and other inappropriate behavior are prohibited in the laboratory. Do not sit on laboratory benches or the floor. Do not use your mobile phone during laboratory sessions. Headphones or earbuds may not be worn while experiments are in progress.
10. Arrive on time and leave only after your work has been completed. Guests and children are not permitted in the laboratory.
11. Come to the laboratory prepared to perform the experiment. Read the procedure in advance and understand what you will be doing. Lack of preparation wastes time and is a major cause of laboratory accidents. Know the hazards before beginning the experiment.
12. Missed laboratory sessions may only be made up with permission from the Faculty Laboratory Coordinator.
13. Perform experiments exactly as directed. Do not perform any procedure that is not part of the assigned experiment.
14. Wash your hands thoroughly before leaving the laboratory or immediately after leaving.

Chemical Hygiene & Good Laboratory Practices

1. Never work alone in the laboratory. Perform experiments only when your teaching assistant or instructor is present.
2. Assume that every chemical presents a potential hazard. Read the experimental procedure carefully and understand the hazards before beginning any work.
3. Label all reaction vessels, beakers, flasks, and sample containers with their contents whenever they will be left unattended. Include your name and the date whenever practical.
4. Never leave an active reaction, heating apparatus, or other hazardous operation unattended unless specifically instructed to do so by your teaching assistant or instructor.
5. Keep your work area clean, organized, and uncluttered. Return chemicals and equipment to their proper locations when they are no longer needed.
6. Inspect glassware and equipment before use. Do not use chipped, cracked, or damaged glassware. Report damaged equipment immediately to your teaching assistant or instructor.
7. Avoid direct contact with chemicals. Never pipette by mouth. Use only pipette bulbs or other approved mechanical pipetting devices.

8. Minimize exposure to chemical vapors, dusts, and aerosols. Perform procedures that generate hazardous vapors, gases, or dusts in a properly functioning chemical fume hood.
9. Keep laboratory notebooks, backpacks, coats, and other personal belongings away from chemicals and designated work areas.
10. If you are uncertain about any procedure or the safe handling of a chemical, stop and ask your teaching assistant or instructor before proceeding.

Laboratory Environment Procedures & Safety

1. Do not enter the laboratory unless your teaching assistant or instructor is present. Never work in the laboratory without the supervision of your teaching assistant or instructor.
2. Learn the locations and proper operation of all safety equipment, including eyewash stations, safety showers, fire extinguishers, fire blankets, spill kits, emergency exits, sinks, and first aid supplies.
3. Report all accidents, injuries, chemical spills, equipment failures, and near misses to your teaching assistant or instructor immediately.
4. If you feel faint or ill, sit down immediately and notify your teaching assistant or instructor.
5. If a chemical enters your eye, flush the affected eye with water for at least 15 minutes. Call for assistance immediately and notify your teaching assistant or instructor.
6. If chemicals are spilled on your skin, flush the affected area with large amounts of water for at least 15 minutes unless otherwise directed by the Safety Data Sheet (SDS). Notify your teaching assistant or instructor immediately.
7. Burns should be treated immediately by placing the burned area under cold running water for several minutes. Have someone notify your teaching assistant or instructor. Prompt cooling can significantly reduce pain and blistering.
8. Treat all chemicals with great care. Many are reactive, toxic, corrosive, flammable, or carcinogenic. Never taste a chemical. Smell chemicals only when specifically instructed to do so by gently wafting vapors toward your nose with your hand. Ask your teaching assistant or instructor whenever you have questions about laboratory safety.
9. Clean up all spills immediately, including water spills. Notify your teaching assistant or instructor of any spill before beginning cleanup if the spill involves hazardous materials.
10. Read chemical labels carefully. Read each label three times: when you pick up the container, before using the chemical, and again before returning the container. Many laboratory mistakes—including some dangerous ones—result from using the wrong chemical.
11. Dispose of chemicals properly. Nothing should be poured down the drain unless specifically instructed by your teaching assistant or instructor. Designated waste containers are provided for chemical waste. Dispose of broken glass only in containers specifically marked for broken glass.
12. Place all unused reagents into the properly labeled waste container at the end of the laboratory period. Never return unused reagents to the original reagent bottle. Do not contaminate reagent bottles. Never pipette directly from reagent bottles. Remove only the quantity of reagent that you actually need.
13. Use a chemical fume hood whenever required by the procedure or instructed by your teaching assistant or instructor.
14. Always turn off Bunsen burners when they are not in use. Never leave a burner unattended. Turn off hot plates, heating mantles, and other electrical equipment when they are no longer needed, and verify that they are turned off before leaving the laboratory.
15. Clean your laboratory bench thoroughly and return all equipment and reagents to their designated locations before leaving the laboratory.

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CHAPTER OVERVIEW

3: Technical Guides

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- [3.2: Recording an ATR-IR Spectrum](#)
- [3.3: Measuring Magnetic Susceptibility](#)
- [3.4: Vacuum Filtration Techniques](#)
- [3.5: Recrystallization Techniques](#)
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3.1: Recording an NMR Spectrum

Nuclear Magnetic Resonance Spectroscopy (NMR) can be used to deduce the connectivity of atoms in molecules. Mostly used to analyze hydrogen and carbon, NMR can also be used to analyze a wide variety of elements, including phosphorus, fluorine, nitrogen, and boron. Specific isotopes of these elements are NMR active.

NMR samples are dissolved in deuterated solvents such as CDCl_3 or D_2O , which contain deuterium (^2H) instead of protium (^1H). This is done so signals from the solvent do not overwhelm the ^1H NMR spectrum. The deuterium also allows the instrument to lock the magnetic field, which helps maintain stable conditions during data collection.

To prepare your sample for NMR analysis, obtain a small vial. This will be used to mix your sample with the appropriate solvent. For liquid samples, add 3–5 drops of your compound to the vial and mix with no more than 0.5 mL of an appropriate deuterated solvent (usually CDCl_3). Use a clean, dry pipet to transfer your sample to an NMR tube. Cap the NMR tube and wipe off the outside of the tube with a Kimwipe. Either submit your sample to the instructor or TA for analysis or proceed to the instrument room to obtain the spectrum yourself according to the instructor's guidelines.

Before placing the sample in the instrument, make sure the NMR tube is clean and free of cracks. Insert the tube into a spinner and adjust the depth using the provided depth gauge. At this point, the sample tube should be held from the end above the spinner. Do not hold the sample from the spinner itself. Carefully place the spinner and tube into the top of the magnet using the sample loader. Do not drop the sample into the magnet.

Using the instrument software, load the appropriate experiment for the nucleus you wish to observe (for example, ^1H or ^{13}C). Lock the instrument on the deuterated solvent and allow the system to stabilize. Shim the magnet, if necessary, to obtain a sharp signal. Once the instrument is properly locked and shimmed, start the acquisition of the spectrum.

After the experiment is complete, process the spectrum using the instrument software. Apply Fourier transformation, if necessary, adjust the phase and baseline, and reference the chemical shift using the solvent peak or another appropriate reference. Save the spectrum in the designated folder.

When finished, remove your sample from the magnet and return the instrument to its standby state. Never leave your sample in the instrument unless instructed to do so.

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3.2: Recording an ATR-IR Spectrum

Attenuated Total Reflection (ATR) is an infrared (IR) spectroscopy technique. Unlike traditional IR sampling methods, such as Nujol mulls and KBr pellets, ATR is nondestructive and requires only a small amount of sample. In general, infrared light is directed through a special crystal on which the sample has been placed. As the light reflects internally within the crystal, it produces an evanescent wave—a weak electromagnetic field that extends only a very short distance (typically less than a few micrometers) beyond the surface of the crystal into the sample. Although the light does not pass directly through the sample, this evanescent wave interacts with the sample and is absorbed at characteristic frequencies. The modified light then continues through the crystal to the detector, producing an infrared spectrum.

Additional information about ATR spectroscopy can be found at the following websites:

- https://en.wikipedia.org/wiki/Attenuated_total_reflectance
- <https://www.bruker.com/en/products-and-solutions/infrared-and-raman/ft-ir-routine-spectrometer/what-is-ft-ir-spectroscopy/atr-attenuated-total-reflectance.html>

In the inorganic laboratory, you will primarily be analyzing solid samples. Obtaining an ATR-IR spectrum is straightforward. Follow the operating instructions specific to the instrument being used. In general:

1. Turn on the instrument and allow it to warm up, if necessary.
2. Start the instrument software on the connected computer.
3. Ensure that the ATR crystal is clean, then collect a background spectrum.
4. Place a small amount of finely powdered sample on the ATR crystal.
5. Lower the pressure clamp onto the sample to ensure good contact with the crystal. Do not apply excessive force, as this may damage the crystal.
6. Collect the sample spectrum.
7. Raise the pressure clamp and remove the sample.
8. Clean the ATR crystal with a dry cotton ball. If residue remains, lightly dampen the cotton ball with isopropanol or methanol and gently wipe the crystal clean.
9. Clean the underside of the pressure clamp in the same manner, if necessary.
10. Collect a new background spectrum whenever instructed by your instructor or whenever instrument conditions change significantly.
11. Use the software to retrieve and analyze the spectrum.

Notes

- Always wear appropriate gloves and safety goggles while handling samples and cleaning solvents.
- Avoid scratching the ATR crystal. Hard or abrasive samples should be finely powdered before analysis and should not be pressed onto the crystal with excessive force.

Follow the manufacturer's operating instructions and any laboratory-specific procedures for the instrument being used.

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3.3: Measuring Magnetic Susceptibility

Magnetic susceptibility

Magnetic susceptibility is a useful property for distinguishing between diamagnetic and paramagnetic compounds. For paramagnetic compounds, magnetic susceptibility can be used to determine the effective magnetic moment, which provides information about the number of unpaired electrons and the electronic configuration of the metal center. The following procedure describes how to obtain a magnetic susceptibility reading using a Johnson–Matthey Susceptibility Balance and calculate the effective magnetic moment of a compound. After obtaining the balance readings, use the equations below to determine the corrected molar susceptibility, the effective magnetic moment, and the number of unpaired electrons.

Calculate the mass susceptibility:

$$\chi_g = \frac{C_{\text{bal}}, l, (R - R_0)}{(m_{s,t} - m_t) \times 10^9} \quad (3.3.1)$$

Multiply the mass susceptibility by the molar mass of the compound to obtain the molar susceptibility:

$$\chi_M = M\chi_g \quad (3.3.2)$$

Correct the molar susceptibility for diamagnetism using Pascal's constants. A table of the constants used in this manual is provided below and is based on the revised values reported by G. A. Bain and J. F. Berry, *Journal of Chemical Education*, **2008**, *85*, 532. The values in this table are tabulated as ****negative**** susceptibilities. Use the absolute values of the appropriate constants when calculating the diamagnetic correction. Sum the absolute values of the corrections for all atoms or ions in the compound and add the result to the measured molar susceptibility as shown below.

$$\chi_A = \chi_M + \sum_i |\chi_{\text{dia},i}| \quad (3.3.3)$$

Using the corrected molar susceptibility, calculate the effective magnetic moment:

$$\mu_{\text{eff}} = 2.828\sqrt{\chi_A T} \quad (3.3.4)$$

The spin-only magnetic moment expected for a transition-metal ion can be calculated as follows:

$$\mu_{\text{eff}} = \sqrt{n(n+2)} \quad (3.3.5)$$

Compare the experimentally determined magnetic moment with the theoretical spin-only values to estimate the number of unpaired electrons in the compound. For first-row transition-metal complexes, the experimentally determined magnetic moment is often close to the spin-only value. Significant deviations may indicate orbital contributions, magnetic coupling between metal centers, temperature effects, or experimental error.

Variables

χ_g = mass susceptibility, $\text{cm}^3 \text{g}^{-1}$

C_{bal} = balance calibration constant

l = sample length (cm)

R = balance reading with the sample present

R_0 = balance reading for the empty sample tube (blank reading)

$m_{s,t}$ = mass of the sample tube and sample (g)

m_t = mass of the empty sample tube (g)

M = molar mass, g mol^{-1}

χ_M = molar susceptibility, $\text{cm}^3 \text{mol}^{-1}$

$\chi_{\text{dia},i}$ = diamagnetic correction for the i th atom or ion, $\text{cm}^3 \text{mol}^{-1}$

χ_A = corrected molar susceptibility, $\text{cm}^3 \text{mol}^{-1}$

μ_{eff} = effective magnetic moment (BM)

T = temperature (K)

n = number of unpaired electrons

Note

CGS System (Standard: cm^3/g (or emu/g , also written as $\text{erg} \cdot \text{G}^{-2} \cdot \text{g}^{-1}$).

Obtaining Data Using the Johnson Matthey Magnetic Susceptibility Balance

IMPORTANT – The balance is very fragile. In general, it should never be moved without securing the internal mechanism. There is a ribbon inside that can be severely damaged, rendering the balance useless. The balance should be placed in an area free of ferromagnetic or other metallic materials. Traffic around the balance can affect the readings obtained.

These steps should be carried out in order, by one group at a time, without interruption.

1. If it is not already on, turn the range knob to $\times 1$ and allow the balance to warm up for 15–30 minutes.
2. With no tube present, use the zero knob to adjust the reading to 000.
3. Carefully place an empty tube in the balance and record R_0 .
4. Remove the tube and immediately record its mass to the nearest 0.001 g.
5. Put the sample in the pre-weighed tube and tap it until compact. Clean up any spills. You will need 3.0–3.5 cm of sample in the tube for good results.
6. Wipe off the tube with a ChemWipe.
7. Place the packed tube in the balance and record the initial reading, R . Remove the tube and tap it down several times. Place the tube back in the balance and record the second reading. If the two readings are the same, continue to the next step. If they are not the same, tap the sample down further and repeat until at least two duplicate readings are obtained.
8. Remove the tube from the balance and immediately measure the height of the sample in cm.
9. Record the room temperature in K.
10. Record the mass of the packed tube in g.
11. Calculate the mass of the sample in g.
12. Remove the sample from the tube and clean up.

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3.4: Vacuum Filtration Techniques

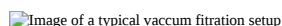
Throughout this laboratory, you will frequently need to collect your product by vacuum filtration. Vacuum filtration is a rapid and efficient method for separating a solid product from a liquid. Unlike gravity filtration, vacuum filtration uses reduced pressure to draw the liquid through a filter, allowing the solid product to be collected more quickly while also removing much of the remaining solvent.

Equipment

A typical vacuum filtration apparatus consists of:

- Büchner funnel
- Filter paper
- Rubber funnel adapter or stopper
- Filter (vacuum) flask with side arm
- Vacuum trap
- Vacuum tubing
- Vacuum source

A labeled diagram of the apparatus should be consulted before assembling the equipment.

Image of a typical vacuum filtration setup

Procedure

Vacuum filtration involves pouring a liquid containing the precipitated product into a Büchner funnel lined with filter paper. The funnel is attached to a filter flask connected to a vacuum source. The vacuum draws the liquid through the filter paper and through the holes in the bottom of the funnel into the filter flask, while the solid product remains on the filter paper.

Before beginning the filtration:

1. Assemble the apparatus and ensure that all tubing connections are secure.
2. Select the correct size filter paper so that it completely covers the holes in the Büchner funnel without extending up the sides.
3. Moisten the filter paper with a small amount of the same solvent used in the filtration. Turn on the vacuum briefly to seat the paper firmly against the funnel and create a good seal.
4. Pour the reaction mixture into the funnel. If necessary, rinse the reaction vessel with small portions of solvent to transfer any remaining product.
5. If instructed, wash the collected solid with small portions of cold solvent to remove impurities.
6. Continue drawing air through the solid for several minutes to remove excess solvent and partially dry the product.

To protect the vacuum pump from accidental contamination, a trap should always be placed between the filter flask and the vacuum source. The trap prevents liquid from being drawn into the vacuum tubing or pump.

Troubleshooting

In some cases, the filter paper may become clogged if the product contains very fine particles relative to the pore size of the filter paper. In these situations, adding a filter aid may improve filtration, or a different grade of filter paper with a more appropriate pore size may be required.

If filtration is unusually slow, check that:

- The filter paper is seated properly.
- The vacuum tubing is securely connected.
- The filter flask and tubing are free of leaks.
- The filter paper is not clogged.

If solid particles pass through the filter paper, the pore size may be too large and a finer grade of filter paper should be used.

Safety

- Always inspect the filter flask for cracks before applying vacuum.
- Wear appropriate personal protective equipment, including safety goggles, a lab coat, and appropriate gloves.
- Secure the apparatus to prevent tipping.
- Never disconnect the vacuum tubing until the vacuum has been released.
- Follow the experimental procedure when selecting the appropriate filtration solvent and washing solvent.

Tips for Success

- Use only enough wash solvent to remove impurities; excessive washing may reduce your yield.
- Whenever possible, use cold wash solvent to minimize dissolution of your product.
- Continue pulling air through the product for several minutes before transferring it to a watch glass or weighing container.
- Record any unusual observations during filtration, such as changes in color, crystal size, or filtration rate, in your laboratory notebook.

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3.5: Recrystallization Techniques

Purpose

Recrystallization is a technique used to purify a solid compound. It is one of the most common methods of purification because many impurities remain dissolved in solution while the desired compound crystallizes. The success of a recrystallization depends on selecting an appropriate solvent or solvent pair and using the minimum amount of solvent necessary.

Principles

Recrystallization purifies a solid because, as crystals form, the molecules of the desired compound arrange themselves into a highly ordered crystal lattice. Most impurities do not fit well into this lattice and therefore remain dissolved in the solvent or are removed during filtration. Successful recrystallization depends on selecting an appropriate solvent or solvent pair and using the minimum amount of solvent necessary.

There are many ways to recrystallize a compound. Two methods will be used in this course.

Single-Solvent Recrystallization

One method involves dissolving the compound in the minimum amount of a hot solvent in which it is highly soluble at elevated temperatures but only sparingly soluble at room temperature. As the solution cools, the solubility of the compound decreases, allowing pure crystals to form while many impurities remain dissolved in the solvent.

Mixed-Solvent Recrystallization

A second method involves using two solvents. The compound is first dissolved in the minimum amount of a solvent in which it is readily soluble. A second solvent, in which the compound is much less soluble but which is completely miscible with the first solvent, is then added slowly until the solution becomes slightly cloudy. Upon cooling, the compound crystallizes while many impurities remain in solution.

Both of these recrystallization methods will be used during this course.

General Procedure

1. Select an appropriate solvent or solvent pair.
2. Place the crude solid in a suitable flask or beaker.
3. Add the minimum amount of solvent required to dissolve the solid.
4. Heat the solution gently, if necessary, until the solid dissolves completely.
5. If insoluble impurities are present, remove them by hot gravity filtration.
6. Allow the solution to cool slowly to room temperature without disturbing it.
7. After crystallization is complete, further cool the solution in an ice bath to maximize crystal formation.
8. Collect the crystals by vacuum filtration.
9. Wash the crystals with a small amount of cold solvent.
10. Allow the crystals to dry thoroughly before determining the mass or performing additional analyses.

Choosing a Recrystallization Solvent

An ideal recrystallization solvent should:

- Dissolve a large amount of the compound when hot.
- Dissolve very little of the compound when cold.
- Leave impurities either completely dissolved or completely insoluble.
- Not react chemically with the compound.
- Be reasonably volatile so it can be removed easily from the crystals.

Tips for Success

- Use the minimum amount of solvent necessary to dissolve the compound.
- Allow the solution to cool slowly. Rapid cooling often produces small crystals that contain more impurities.
- Do not disturb the solution while crystals are forming.

- Use only a small amount of cold solvent to wash the crystals. Excessive washing can reduce the yield.
- If no crystals form, gently scratch the inside of the container with a glass stirring rod or add a seed crystal, if available.

Troubleshooting

- The solid does not dissolve: Add small portions of hot solvent until it just dissolves.
- No crystals form: Cool the solution in an ice bath or gently evaporate some solvent and try again.
- The crystals are very small: Repeat the recrystallization and allow the solution to cool more slowly.
- The crystals appear impure: Repeat the recrystallization.

Safety

- Use caution when heating organic solvents, as many are flammable.
- Never heat flammable solvents over an open flame; use a hot plate or heating mantle as directed.
- Handle hot glassware carefully and allow it to cool before touching.
- Wear appropriate personal protective equipment, including safety goggles, a lab coat, and appropriate gloves.
- Consult the Safety Data Sheet (SDS) for any unfamiliar solvent before beginning the experiment.

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3.6: Cleaning Glassware

Purpose

Clean glassware is essential for successful laboratory work. Residual chemicals or contamination from previous experiments can interfere with chemical reactions, reduce product purity, contaminate analytical measurements, and in some cases prevent a reaction from occurring. Proper cleaning also reduces potential safety hazards and helps maintain laboratory equipment in good condition.

Principles

Glassware should be cleaned as soon as possible after use. Allowing chemicals to dry on the glass surface often makes them much more difficult to remove and may damage the glassware. Before cleaning, all chemical waste should be disposed of in the appropriate waste container according to the laboratory instructions.

Different types of glassware require different cleaning procedures. In the inorganic laboratory, three general categories of glassware are commonly used:

- General laboratory glassware (beakers, Erlenmeyer flasks, test tubes, reaction flasks, watch glasses, etc.)
- Graduated glassware (burets, graduated cylinders, volumetric flasks, pipets, etc.)
- Air-free glassware (Schlenk flasks, Schlenk tubes, and related equipment)

General Procedure

General Laboratory Glassware

1. Empty the contents into the appropriate waste container.
2. Rinse the glassware with small portions of a suitable solvent (for example, acetone) to remove as much residual material as possible.
3. Wash thoroughly with laboratory detergent and warm water using an appropriate brush if necessary.
4. Rinse thoroughly with tap water to remove all detergent.
5. Perform a final rinse with deionized water.
6. Allow the glassware to air dry or dry as directed by your instructor.

Graduated Glassware

Follow the same procedure used for general laboratory glassware. Take care not to scratch the interior surfaces of burets, pipets, or volumetric glassware, as scratches may affect measurement accuracy.

Air-Free Glassware

Follow the same cleaning procedure used for general laboratory glassware. After the final water rinse:

1. Rinse the glassware with acetone to remove residual water.
2. Place the glassware in a drying oven until completely dry.
3. Allow the glassware to cool before use or storage.

Using a Base Bath

Occasionally, glassware may remain contaminated after normal washing. In these cases, a base bath (typically a solution of potassium hydroxide in isopropanol) may be used.

- First rinse away as much chemical residue as possible and wash the glassware with detergent and water.
- Soak the glassware in the base bath only as long as necessary.
- Remove the glassware and rinse thoroughly with tap water.
- If desired, rinse briefly with dilute acid to neutralize any remaining base.
- Finish with a thorough rinse using deionized water and allow the glassware to air dry.

Tips for Success

- Clean glassware immediately after use whenever possible.
- Use only the amount of solvent needed to remove chemical residues.
- Verify that all detergent has been removed before the final rinse.
- Inspect the glassware after cleaning. Properly cleaned glassware should drain evenly without water beading on the surface.
- Replace chipped or cracked glassware rather than attempting to use it.

Troubleshooting

- Residue remains after washing: Repeat the cleaning procedure or use a base bath if appropriate.
- Water beads on the glass: Continue washing until the glass is completely clean.
- Labels or grease remain: Use an appropriate solvent before washing with detergent.
- Glassware still appears dirty: Repeat the cleaning process before using the glassware.

Safety

- Always dispose of chemical waste in the designated waste container before washing glassware.
- Wear appropriate personal protective equipment, including safety goggles, a lab coat, and appropriate gloves.
- Handle broken or chipped glassware carefully and dispose of it in the designated broken-glass container.
- Never leave glassware in a base bath for more than 24 hours.
- Never place fritted glassware in a base bath, as the frit may be damaged or become clogged.
- Avoid prolonged exposure of ground-glass joints to a base bath, as the joints may become etched and difficult to separate.
- Use acetone and other organic solvents only in well-ventilated areas and keep them away from ignition sources.

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CHAPTER OVERVIEW

4: Experiments

- 4.1: Synthesis and Characterization of Potassium Trioxalatoferrate(III)
- 4.2: Redox Titration of Potassium Trioxalatoferrate(III)
- 4.3: Synthesis of Metal Acetylacetonates
- 4.4: Synthesis and Characterization of a Metal Hydride Complex
- 4.5: Ni(II) Coordination Complexes
- 4.6: Preparation of (benzene)tricarbonylchromium(0)
- 4.7: Synthesis of a High Temperature Superconductor
- 4.8: Cobalt Complex of Dioxygen
- 4.9: Synthesis of Some Representative Metal Organic Frameworks (MOFs)

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4.1: Synthesis and Characterization of Potassium Trioxalatoferrate(III)

Learning Objectives

- Synthesize potassium trioxalatoferrate(III) trihydrate, $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$, using common inorganic synthesis techniques, including precipitation, filtration, recrystallization, and crystal isolation.
- Describe the structure and bonding of the trioxalatoferrate(III) complex, including the chelating behavior of oxalate.
- Characterize the synthesized complex using physical observations, percent yield, and infrared spectroscopy.
- Determine the oxalate content of potassium trioxalatoferrate(III) by redox titration and calculate the experimental oxalate-to-iron ratio. (See Exp 4.2)
- Assess the purity and composition of the product by comparing experimental and theoretical results and considering sources of error.

Introduction

The potassium salt of trioxalatoferrate(III) is a classic example of a coordination compound. Also known as tris(oxalato)ferrate(III), the trivalent anion has the formula $[Fe(C_2O_4)_3]^{3-}$. The complex ion contains an iron atom in the +3 oxidation state coordinated to three bidentate oxalate ligands.

$K_3[Fe(C_2O_4)_3]$ is very light-sensitive and will decompose through a photoredox reaction if exposed to light for extended periods. During this process, the iron(III) center is reduced to iron(II), while the oxalate ligand is oxidized to carbon dioxide. This property makes $K_3[Fe(C_2O_4)_3]$ useful for chemical actinometry, the measurement of light intensity.

Historically, potassium ferrioxalate was used in alternative photographic contact-printing processes. These methods involved coating paper with the complex and exposing it to light through a photographic negative or stencil. Areas exposed to light underwent reduction of iron(III) to iron(II). The resulting iron(II) species was then treated with potassium ferricyanide to produce intensely blue images with the insoluble pigment known as Prussian blue.

$K_3[Fe(C_2O_4)_3]$ has also been used industrially in photo-Fenton water treatment systems. Iron(II) generated photochemically reacts with hydrogen peroxide to produce highly reactive hydroxyl radicals that degrade pharmaceuticals, dyes, and other organic contaminants in wastewater.

In addition, $K_3[Fe(C_2O_4)_3]$ serves as a catalyst precursor for the synthesis of nanoscale ferrite materials such as $NiFe_2O_4$ and $CuFe_2O_4$.

In this experiment, you will synthesize and characterize potassium trioxalatoferrate(III).

Experimental Method

Safety Precautions

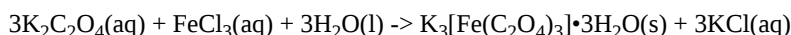
- Wear appropriate PPE at all times. Safety goggles, a lab coat, and gloves should be worn throughout the experiment to protect against chemical splashes and spills.
- Avoid skin and eye contact with iron(III) chloride solutions. Iron(III) chloride is corrosive and can cause irritation or burns upon contact.
- Handle potassium oxalate carefully. Potassium oxalate is toxic if ingested and may be harmful if inhaled or absorbed through prolonged skin contact.
- Exercise caution when heating solutions. Use care when working with hot glassware and solutions to prevent burns and injuries from splashing.
- Dispose of chemical waste properly. All solutions containing iron, oxalate, or permanganate ions should be placed in the designated waste container.
- Handle potassium permanganate with care during the titration. Potassium permanganate is a strong oxidizing agent that can stain skin and clothing and should be kept away from combustible materials.

Materials

Chemical Name	Formula	CAS Number	Molar Mass (g/mol)	Density (g/mL)
Iron(III) chloride hexahydrate	$FeCl_3 \cdot 6H_2O$	10025-77-1	270.30	— (solid)

Potassium oxalate monohydrate	$\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$	6487-48-5	184.23	— (solid)
Acetone	$\text{C}_3\text{H}_6\text{O}$	67-64-1	58.08	0.785
Water	H_2O	7732-18-5	18.02	0.998

Procedure



1. Accurately weigh 9.000 g of potassium oxalate monohydrate, recording the mass to the nearest 0.001 g.
2. Transfer the solid to a clean, dry 100 mL beaker and add 22 mL of deionized water. Place the beaker on a hot plate and heat gently, stirring occasionally, until the solid dissolves completely. Do not allow the solution to boil.
3. Obtain approximately 1.6 g of iron(III) chloride hexahydrate in a clean, dry, pre-weighed 50 mL beaker. Determine the exact mass of the iron(III) chloride by difference, and record the mass to the nearest 0.001 g.
4. After the potassium oxalate monohydrate has dissolved, carefully pour the hot solution into the beaker containing the iron(III) chloride hexahydrate. Record any observations in your lab notebook. Stir the mixture with a clean glass stirring rod until the iron(III) chloride dissolves completely.
5. Place the 50 mL beaker in an ice bath and allow the solution to cool for 10 minutes, stirring occasionally during this period.
6. If no solid product has formed after 10 minutes, gently induce crystallization by scratching the bottom of the beaker with a clean glass stirring rod. Return the beaker to the ice bath and allow the mixture to stand for an additional 5 minutes.
7. Isolate the crystals using a Büchner funnel. **IMPORTANT:** Do not rinse the 50 mL beaker, even if you are unable to transfer all of the crystals to the funnel. Pour as much of the mixture as possible into the Büchner funnel, but retain the beaker for the next step. Record the appearance (color and clarity) of the filtrate. Immediately transfer the isolated crystals back to the original beaker containing any crystals that were not collected during filtration.
8. Carefully rinse any crystals adhering to the sides of the beaker to the bottom using the minimum volume of distilled water necessary. Do not use more than 10 mL.
9. Dissolve the crystals completely by gently heating the beaker on a hot plate. Do not allow the solution to boil.
10. Once the crystals have dissolved, carefully move the beaker to a cool, dark location and allow it to stand undisturbed for 15–20 minutes. If crystals have formed after 20 minutes, place the beaker in an ice bath for 5 minutes. Record the appearance of both the crystals and the mother liquor. If no crystals have formed after 20 minutes, consult the instructor.
11. After successfully completing Step 10, isolate the crystals by suction filtration. Rinse any remaining crystals from the beaker into the Büchner funnel using a small portion of cold deionized water.
12. Carefully rinse the crystals with a small portion of acetone. Transfer the crystals to a vial and cover the opening with a tissue (e.g., a Kimwipe®) secured with a rubber band. Place the vial in a desiccator and allow the crystals to dry until the next laboratory period. The product is light-sensitive. Keep the desiccator covered or store it in a dark location between laboratory sessions.

Data Collection and Analysis

1. Describe the color of the product
2. Calculate the % yield. The calculation must be in your laboratory notebook.
3. Collect, analyze and interpret an ATR-IR spectrum. Discuss what the observed peaks or peak patterns reveal in each spectrum. You may reference literature sources to support your interpretation.
4. Carry out a redox titration and calculate the experimental oxalate-to-iron ratio. (See Exp 4.2)

Questions and Further Thoughts

1. Were any of the reactants used in excess? If so, why?
2. Why was the product recrystallized after the initial isolation.
3. The product is highly light sensitive. Explain the photochemical reaction that occurs upon exposure to light. What is the fate of the oxalate ligand?
4. The oxalate ligand is a bidentate ligand. What does this mean

Notes

Further Readings

1. Kmet', A.; Drozdíková, A.; Nagyová, S.; Ikhardt, P. The Cyanotype Process and Its Potential in Chemistry Education. *J. Chem. Educ.* **2023**, *100* (6), 2367–2372. <https://doi.org/10.1021/acs.jchemed.2c01089>.
2. Junk, P. C.; Bacsa, J.; Thomas, N. C. Growing a Display Crystal of Recycled Potassium Tris(Oxalato)Ferrate(III) Trihydrate. *J. Chem. Educ.* **2021**, *98* (10), 3400–3404. <https://doi.org/10.1021/acs.jchemed.1c00776>.
3. Fiorito, P. A.; Polo, A. S. A New Approach toward Cyanotype Photography Using Tris-(Oxalato)Ferrate(III): An Integrated Experiment. *J. Chem. Educ.* **2015**, *92* (10), 1721–1724. <https://doi.org/10.1021/ed500809n>.
4. González, G.; Seco, M. Potassium Tris(Oxalato)Ferrate(III): A Versatile Compound To Illustrate the Principles of Chemical Equilibria. *J. Chem. Educ.* **2004**, *81* (8), 1193. <https://doi.org/10.1021/ed081p1193>.
5. Dallinger, R. F. Synthesis and Characterization of Potassium Tris(Oxalato)Ferrate(III) Trihydrate: A Spectrophotometric Method of Iron Analysis. *J. Chem. Educ.* **1995**, *72* (10), 936. <https://doi.org/10.1021/ed072p936>.
6. Young, C. G. Tris(Oxalato) Teaching Aids. *J. Chem. Educ.* **1985**, *62* (5), 445. <https://doi.org/10.1021/ed062p445>.
7. Olmsted, J. Preparation and Analysis of Potassium Tris(Oxalato)Ferrate(III)Trihydrate: A General Chemistry Experiment. *J. Chem. Educ.* **1984**, *61* (12), 1098. <https://doi.org/10.1021/ed061p1098>.

Techniques: recrystallization, filtration, ATR-IR

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4.2: Redox Titration of Potassium Trioxalatoferrate(III)

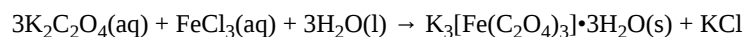
Learning Objectives

- Determine the oxalate content of potassium trioxalatoferrate(III) by redox titration and calculate the experimental oxalate-to-iron ratio.

Introduction

The purpose of this activity is to confirm the ligand stoichiometry for the reaction to prepare $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}(\text{s})$.

During a previous lab session, you prepared potassium trioxalatoferrate(III) by reaction of potassium oxalate with iron(III) chloride according to the following equation.



The oxalate ion, $\text{C}_2\text{O}_4^{2-}$ reacts with the permanganate ion MnO_4^- to form CO_2 and H_2O . We can use this property to determine the ligand stoichiometry for the $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}(\text{s})$ product.

This lab involves 2 major steps.

- Standardize the stock KMnO_4 solution.
- Titrate the $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ sample (at least two trials).

Before you begin, Write a balanced redox equation for the reaction of MnO_4^- with $\text{C}_2\text{O}_4^{2-}$ in acidic solution to form carbon dioxide and $\text{Mn}^{2+}(\text{aq})$ in your notebook. Have your equation checked by the instructor or TA.

Experimental Method

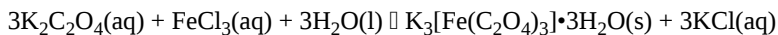
Safety Precautions

- Wear appropriate PPE at all times. Safety goggles, a lab coat, and gloves should be worn throughout the experiment to protect against chemical splashes and spills.
- Handle potassium oxalate carefully. Potassium oxalate is toxic if ingested and may be harmful if inhaled or absorbed through prolonged skin contact.
- Exercise caution when heating solutions. Use care when working with hot glassware and solutions to prevent burns and injuries from splashing.
- Dispose of chemical waste properly. All solutions containing iron, oxalate, or permanganate ions should be placed in the designated waste container.
- Handle potassium permanganate with care during the titration. Potassium permanganate is a strong oxidizing agent that can stain skin and clothing and should be kept away from combustible materials.

Materials

Chemical Name	Formula	CAS Number	Molar Mass (g/mol)	Density (g/mL)
Potassium oxalate monohydrate	$\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$	6487-48-5	184.23	— (solid)
Potassium permanganate (used by TA or instructor to prepare a 0.1 M stock solution)	KMnO_4	7722-64-7	158.03	— (solid)
Sulfuric acid (concentrated)	H_2SO_4	7664-93-9	98.08	1.84
Phosphoric acid (85 wt% aqueous)	H_3PO_4	7664-38-2	98.00	1.69
Water	H_2O	7732-18-5	18.02	0.998

Procedure



Standardizing the KMnO_4 solution.

1. If you haven't done so already determine the mass of your $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ sample and calculate the percent yield. Record the percent yield in your notebook.
2. Weight out two samples of potassium oxalate monohydrate, $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$. Each sample should be ~0.1 to 0.11 g. Record the masses to the nearest milligram (0.001g) in your notebook.
3. Put each sample into separate 125 mL Erlenmeyer flasks. Label the flasks so you can tell them apart. Record how you labeled the flasks in your notebook.
4. Prepare a table in your notebook like the following.

	Flask A	Flask B
Mass of potassium oxalate monohydrate (g)		
Molar Mass (g/mol)		
Moles of potassium oxalate monohydrate (g)		
Moles of KMnO_4 (mol)		
Initial Volume of KMnO_4 (mL)		
Final Volume of KMnO_4 (mL)		
Total Volume added (mL)		
Total Volume added (L)		
Molarity of KMnO_4 (mol/L)		
Average Molarity		

Caution

WEAR GLOVES FOR THE NEXT STEPS.

HANDLE THE HOT AND ACIDIC SOLUTIONS WITH CARE!

5. Add ~35 mL of 1.0 M H_2SO_4 to each flask.
6. Add 5 drops of phosphoric acid to each flask.
7. Heat the flasks on a hot plate set to medium. Periodically swirl the samples to ensure that all the solid dissolves. Heat the solutions until they reach 70°C . **Do not allow the temperature to rise above 80°C .** While the samples are heating, set up a buret following the next steps but keep an eye on your samples. Turn down the hot plate if the temperature rises too quickly.
8. Prepare 80 mL of ~0.02 M KMnO_4 by diluting the 0.1 M stock solution with DI water. Use $M_1V_1 = M_2V_2$ to calculate the volume of KMnO_4 required. Be sure to do the calculations in your notebook. Determine and record the amount of water needed by difference. At this point, you do not need to make this an exact solution. (i.e. volumetric glassware is not needed)

Next, standardize the solution prepared in step 8 to determine its "exact" concentration which should be close to 0.02

9. Rinse your buret with about 10 mL of your KMnO_4 solution.
10. Fill the buret to the 0.00 mL mark. Flush any bubbles from the tip and refill if needed.

11. Record the initial volume of the buret. This may not be 0.00. Be sure to record the correct number of significant figures.
12. Titrate the first hot solution of potassium oxalate by quickly adding about 5 mL of permanganate solution to the sample. After the first 5 mL, add the permanganate solution slowly until a light pink color is observed in the flask. When the light pink color persists for at least 15 seconds, you have reached the endpoint. Record the volume needed to reach the endpoint.
13. Titrate the second sample. You may do the second sample more quickly by adding permanganate solution to within 2 mL of the volume added for the first sample, then add permanganate slowly to reach the endpoint.
14. Calculate the relative percent difference (RPD) in the volumes required to reach the end point. $RPD = |V_1 - V_2| / (V_1 + V_2) / 2 \times 100$. If the RPD is $> 2\%$, consider doing a third determination before proceeding.
15. Calculate the molarity of your standardized $KMnO_4$ solution. Use the average of the two determinations.

Titration of the Potassium Trioxalatoferrate(III) trihydrate

16. Refill the buret with your standardized solution for the next titrations.
17. Weigh two samples of your $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$. Each sample should have a mass of 0.1 to 0.11 g. Record the mass to the nearest milligram in your notebook.
18. Prepare the samples as you did before and titrate the hot solutions to the endpoint. Be sure to record the initial and final volumes of permanganate solution used for each sample.

	Flask A	Flask B
Mass of $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$ (g)		
Initial Volume of $KMnO_4$ (mL)		
Final Volume of $KMnO_4$ (mL)		
Total Volume added (mL)		
Total Volume added (L)		
Standardized Molarity		
Moles of $KMnO_4$ added		
Moles of oxalate		
Molar Mass of $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$ (g/mol)		
Moles of sample		
Ratio of moles of oxalate to moles of sample		

Data Collection and Analysis

1. Calculate and report the molarity of the standardized $KMnO_4$.
2. Using the balanced redox equation for the reaction of MnO_4^- with $C_2O_4^{2-}$ in acidic solution, calculate moles of $C_2O_4^{2-}$ that reacted in each sample.
3. Determine the ratio of moles of $C_2O_4^{2-}$ to moles of sample.

Questions and Further Thoughts

1. Explain why the permanganate titration can be used to verify the 3:1 oxalate-to-iron ratio in $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$ and assess the purity of the product.

2. Why is the titration performed in hot, acidic solution? How would insufficient heating or acidity affect the results?
3. If your experimental oxalate-to-iron ratio is less than the theoretical value of 3.00, suggest three possible reasons for the discrepancy.

Notes

Clean up. Put all chemical waste into the appropriate waste container. Wash all glassware and return it to your hood to dry for next week.

Further Readings

González, G.; Seco, M. Potassium Tris(Oxalato)Ferrate(III): A Versatile Compound To Illustrate the Principles of Chemical Equilibria. *J. Chem. Educ.* **2004**, 81 (8), 1193. <https://doi.org/10.1021/ed081p1193>.

Dallinger, R. F. Synthesis and Characterization of Potassium Tris(Oxalato)Ferrate(III) Trihydrate: A Spectrophotometric Method of Iron Analysis. *J. Chem. Educ.* **1995**, 72 (10), 936. <https://doi.org/10.1021/ed072p936>.

Young, C. G. Tris(Oxalato) Teaching Aids. *J. Chem. Educ.* **1985**, 62 (5), 445. <https://doi.org/10.1021/ed062p445>.

Olmsted, J. Preparation and Analysis of Potassium Tris(Oxalato)Ferrate(III)Trihydrate: A General Chemistry Experiment. *J. Chem. Educ.* **1984**, 61 (12), 1098. <https://doi.org/10.1021/ed061p1098>.

Techniques: redox titration,

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4.3: Synthesis of Metal Acetylacetonates

Learning Objectives

- Explain keto–enol tautomerism in β -diketones and describe how deprotonation leads to formation of the acetylacetonate ligand (acac^-).
- Describe the coordination behavior of the acetylacetonate ligand, including its role as a bidentate chelating ligand that forms six-membered chelate rings with metal ions.
- Synthesize a metal acetylacetonate complex using appropriate laboratory techniques such as controlled addition of reagents, precipitation, filtration, and drying.
- Relate metal oxidation state and electron configuration to magnetic properties of coordination complexes.
- Determine the number of unpaired electrons in a metal complex using magnetic susceptibility measurements.
- Interpret the relationship between electronic structure and observed color in transition-metal acetylacetonate complexes.
- Evaluate experimental results by comparing measured magnetic data with theoretical predictions based on electron configurations.

Introduction

β -Diketones such as acetylacetone (2,4-pentanedione) exist in equilibrium between two forms: the keto form and the enol form. When a base is added, acetylacetone can be deprotonated (from the enol form) to form the acetylacetonate anion (acac^-). This anion can act as a bidentate ligand when coordinating to metal ions.

The acetylacetonate ligand binds through two oxygen atoms, forming a six-membered chelate ring with the metal center. Depending on the oxidation state and coordination environment of the metal, several acac ligands may bind to a single metal ion.

Metal acetylacetonate complexes display a variety of colors depending on the metal involved. These complexes also have practical applications in areas such as catalysis and materials synthesis. The metals involved may have different numbers of unpaired electrons, making some of these complexes diamagnetic while others are paramagnetic.

In this experiment, you will prepare at least one metal acetylacetonate complex and determine, by magnetic susceptibility, the number of unpaired electrons.

Experimental Method

Safety Precautions

- Methanol is toxic and can be absorbed through the skin or inhaled. Avoid contact and inhalation of vapors.
- Acetylacetone is flammable and irritating. Keep away from ignition sources and avoid inhalation of vapors.
- Sulfuric acid is highly corrosive, and ammonium hydroxide and sodium carbonate are basic. Add acids and bases slowly with stirring to avoid splashing.
- Hydrogen peroxide is a strong oxidizing agent and should be added slowly with stirring.
- Many transition-metal salts (Cr, Co, V, Cu, Fe, Mn) are harmful if ingested or inhaled. Avoid contact with skin and wash hands after handling.
- Several procedures require heating solutions to 80–90 °C. Use caution when working with hot plates and hot glassware.

Materials

Chemical Name	Formula	CAS Number	Molar Mass (g/mol)	Density (g/mL or g/cm ³)
Aluminum Sulfate	$\text{Al}_2(\text{SO}_4)_3$	10043-01-3	342.15	2.71
Vanadium Pentoxide	V_2O_5	1314-62-1	181.88	3.36
Chromium(III) Chloride Hexahydrate	$\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$	10060-12-5	266.45	1.76
Manganese(II) Chloride Tetrahydrate	$\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$	13446-34-9	197.91	2.01

Iron(II) Tetrahydrate	Chloride	$\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$	13478-10-9	198.81	1.93
Cobalt Carbonate		CoCO_3	513-79-1	118.94	4.13
Copper(II) Dihydrate	Chloride	$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$	10125-13-0	170.48	2.51
Deionized Water		H_2O	7732-18-5	18.02	1.00
Methanol		CH_3OH	67-56-1	32.04	0.792
Ethanol		$\text{C}_2\text{H}_5\text{OH}$	64-17-5	46.07	0.789
Acetylacetone		$\text{C}_5\text{H}_8\text{O}_2$	123-54-6	100.12	0.975
Sulfuric Acid (conc.)		H_2SO_4	7664-93-9	98.08	1.84
Ammonia (dilute)	Solution	$\text{NH}_3(\text{aq})$	1336-21-6	17.03 (NH_3)	Variable*
Hydrogen (10%)	Peroxide	$\text{H}_2\text{O}_2(\text{aq})$	7722-84-1	34.01 (H_2O_2)	1.03
Sodium Acetate (anh.)		CH_3COONa	127-09-3	82.03	1.53
Sodium Bicarbonate		NaHCO_3	144-55-8	84.01	2.20
Urea		$\text{CO}(\text{NH}_2)_2$	57-13-6	60.06	1.32

Density of ammonia solutions depends on concentration. A typical dilute laboratory ammonia solution has a density near 0.96–0.99 g/mL.

Procedure

Preparation of Metal Acetylacetonates

Al^{3+} complex



1. Add 3 mL of acetylacetone and 40 mL of distilled water to a 100 mL Erlenmeyer flask.
2. Add 8 mL of 5 M ammonium hydroxide and swirl to mix.
3. In a separate flask, dissolve 3.0 g of aluminum sulfate [$\text{Al}_2(\text{SO}_4)_3 \cdot 16\text{H}_2\text{O}$] in 30 mL of cold distilled water.
4. While stirring, slowly add the ammoniacal acetylacetone solution to the aluminum sulfate solution.
5. Check the pH using pH paper and adjust to neutral with 5 M ammonium hydroxide if necessary.
6. Allow the mixture to stand for 10 minutes. After 5 minutes, place the flask in an ice bath if needed to complete precipitation.
7. Collect the product by Büchner filtration.
8. Wash the product with cold distilled water (3×20 mL).
9. Allow the solid to air-dry for approximately 10 minutes.
10. Dry the product in a vacuum desiccator.
11. Determine the percent yield.
12. Recrystallize the product from cyclohexane or petroleum ether.
13. Determine the melting point of the purified product.

$(\text{VO})^{2+}$ complex

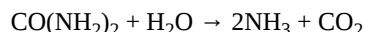


1. Add 5 mL of deionized water to a 50 mL round-bottom flask.
2. Slowly add 5 mL of concentrated H_2SO_4 to the flask.

3. Add 12 mL of ethanol.
4. Add 2.5 g (0.014 mol) of V_2O_5 .
5. Attach the flask to a water-cooled condenser.
6. Reflux the mixture for 1.5 hours.
7. The solution should gradually turn a dark blue-green color.
8. Cool the flask under running tap water, then filter the solution through cotton wool to remove any remaining solid material.
9. Add 6 mL of acetylacetone dropwise to the filtrate with stirring.
10. Prepare a sodium carbonate solution by dissolving 20 g of anhydrous Na_2CO_3 in 150 mL of distilled water in a 500 mL Erlenmeyer flask with stirring.
11. Neutralize the reaction mixture by carefully adding it to the sodium carbonate solution with stirring.
12. Cool the flask in an ice bath for at least 15 minutes.
13. Collect the dark green product by Büchner filtration.
14. Wash the solid with cold distilled water (2×15 mL).
15. Allow the product to air-dry for at least 15 minutes.
16. Transfer the product to a vacuum desiccator for final drying.
17. Weigh the dried product and calculate the percent yield.
18. Recrystallize the product by dissolving a portion in the minimum amount of dichloromethane (approximately 500 mg in 6 mL).
19. Decant the solution from any insoluble residue.
20. Add 20 mL of petroleum ether and allow the solution to stand for 10 minutes.
21. Collect the product by Büchner filtration.
22. Dry the product and record the melting point.

Note: Ethanol reacts with V_2O_5 in concentrated sulfuric acid to produce vanadyl sulfate ($VOSO_4$), acetaldehyde (CH_3CHO), and water. The vanadyl ion can then coordinate with acetylacetonate ligands to form a vanadyl acetylacetonate complex.

Cr³⁺ complex



1. Weigh 1.4 g of chromium(III) chloride hexahydrate ($CrCl_3 \cdot 6H_2O$) directly into a 100 mL Erlenmeyer flask.
2. Add 50 mL of distilled water and swirl until the chromium(III) chloride hexahydrate dissolves.
3. Weigh 10 g of urea and add it to the deep green chromium solution in 3–4 portions with stirring.
4. Using a pipette, add 3 mL of acetylacetone to the chromium solution.
5. Cover the reaction vessel with a watch glass and heat the mixture to 80–90 °C on a hot plate. Carefully monitor and maintain the temperature using a short-stem thermometer. Do not overheat the solution. Initially, the solution should appear very dark; however, as the reaction proceeds, maroon-colored crystals will form.
6. Heat the reaction mixture for 1.5 hours, then allow it to cool to room temperature.
7. Collect the product by Büchner filtration. Do not wash the product. Allow it to air-dry.
8. Determine the mass of the product and calculate the percent yield.
9. Dissolve a small sample of the product in boiling cyclohexane or petroleum ether. Decant the solution from any insoluble material, reduce the volume to approximately one-half, and then allow the solution to cool.
10. Collect the deep red needles by Büchner filtration.
11. Allow the product to dry, then determine its melting point.

Note: The ammonia liberated by the slow hydrolysis of urea helps control the pH of the reaction mixture.

Mn³⁺ complex



1. Dissolve 2.6 g of manganese chloride ($MnCl_2 \cdot 4H_2O$) and 6.8 g of sodium acetate in 100 mL of distilled water.
2. Add 10 mL of acetylacetone to the manganese chloride solution.
3. In a separate flask, prepare a solution of potassium permanganate by dissolving 0.52 g of $KMnO_4$ in 25 mL of distilled water. Stir the mixture to ensure that all of the potassium permanganate dissolves before proceeding.

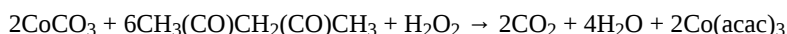
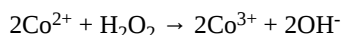
- Over a period of 10–15 min, add the potassium permanganate solution to the solution prepared in Steps 1–2.
- Stir for an additional 10 min. Then, over a 10 min period, add a solution of sodium acetate prepared by dissolving 6.3 g of sodium acetate in 25 mL of distilled water.
- While continuing to stir, heat the flask on a hot plate to 60–70 °C for 15 min.
- Allow the flask to cool, then place it in an ice bath.
- Using Büchner filtration, isolate the dark precipitate and wash it with small portions of cold distilled water.
- Dry the product in a vacuum desiccator.
- Determine the percent yield of the dry product.
- Recrystallize the product from warm petroleum ether or cyclohexane.
- Use Büchner filtration to isolate the black crystalline product.
- Rinse the product with a small amount of cold petroleum ether and transfer it to a vacuum desiccator for final drying.

Fe³⁺ complex



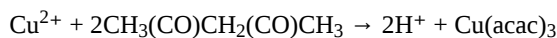
- Dissolve 3.3 g of iron(II) chloride (FeCl₂·6H₂O) in 25 mL distilled water.
- With stirring add a solution of acetylacetone in 10 mL methanol.
- Add to the mixture a solution of 5.1 g sodium acetate in 15 mL distilled water. At this point a red precipitate should form.
- Heat the reaction to 80 °C for 15 minutes.
- Remove the flask from the heat and allow it to cool.
- Place the cool flask in an ice bath.
- Using Büchner filtration, isolate the precipitate and wash it with small portions of cold distilled water.
- Dry the product in a vacuum desiccator.
- Determine the percent yield of the dry product.
- Recrystallize the product from warm methanol.
- Use Büchner filtration to isolate the red crystalline product.
- Rinse the product with a small amount of cold methanol and transfer it to a vacuum desiccator for final drying.

Co³⁺ complex



- In a 100 mL Erlenmeyer flask, dissolve 2.5 g of cobalt carbonate (CoCO₃) in 20 mL acetylacetone.
- With continuous stirring, heat the mixture to about 90 °C.
- Over a 30-minute period while monitoring and maintaining the temperature at 90 °C, add 30 mL of a 10% H₂O₂ solution dropwise. Cover the flask with a watch glass between additions.
- Continue to heat the reaction at 90 °C for 15 more minutes.
- Remove the flask from the heat and allow it to cool.
- Place the cooled flask in an ice bath.
- Using Büchner filtration, isolate the dark green precipitate.
- Dry the product in an oven at 100 °C.
- Determine the percent yield of the dry product.
- Recrystallize the product from petroleum ether.
- Use Büchner filtration to isolate the dark green crystalline product.
- Rinse the product with a small amount of cold petroleum ether and transfer it to a vacuum desiccator for final drying.

Cu²⁺ complex



- In an Erlenmeyer flask dissolve 4 g of copper(II) chloride (CuCl₂·6H₂O) in 25 mL distilled water.
- Over a period of 10 minutes with stirring add 5 mL acetylacetone in 10 mL methanol.
- Over a 5 min period add to the mixture a solution of 6.8 g sodium acetate dissolved in 15 mL distilled water.
- Heat the reaction to 80 °C for 15 minutes.

5. Remove the flask from the heat and allow it to cool.
6. Place the cooled flask in an ice bath.
7. Using Büchner filtration, isolate the blue-grey precipitate and wash it with small portions of cold distilled water.
8. Dry the product in an oven at 100 °C.
9. Determine the percent yield of the dry product.
10. Recrystallize the product from methanol.
11. Use Büchner filtration to isolate the blue-grey crystalline product.
12. Rinse the product with a small amount of cold methanol and transfer it to a vacuum desiccator for final drying.

Data Collection and Analysis

1. Describe the color of the product.
2. Calculate the % yield. The calculation must be in your laboratory notebook.
3. Collect magnetic susceptibility data.
4. Collect, analyze and interpret an ATR-IR spectrum. Discuss what the observed peaks or peak patterns reveal in each spectrum. You may reference literature sources to support your interpretation.

Questions and Further Thoughts

1. Why must acetylacetone be deprotonated before it can coordinate to a metal ion?
2. Why is acetylacetonate considered a bidentate ligand?
3. What is the chelate effect, and why does it make metal acetylacetonate complexes more stable than analogous complexes containing monodentate ligands?
4. How does the identity of the metal ion influence the color and magnetic properties of its acetylacetonate complex?
5. Suggest one source of experimental error that could affect the percent yield or purity of your product, and explain its effect.

Notes

Further Readings

1. Glidewell, C. *Metal Acetylacetonate Complexes: Preparation and Characterisation*. In *Inorganic Experiments*; 3rd ed.; Dale B. Woollins, Ed.; Wiley-VCH: Weinheim, Germany, 2010; pp. 109–119.
2. Seco, M. Acetylacetone: A Versatile Ligand. *J. Chem. Educ.* **1989**, 66 (9), 779. <https://doi.org/10.1021/ed066p779>.
3. Greenaway, A. M.; Trail, L. E. Magnetic Susceptibility: A Practical Introduction. *J. Chem. Educ.* **1983**, 60 (8), 681. <https://doi.org/10.1021/ed060p681>.

Techniques: filtration, reflux, recrystallization, magnetic susceptibility, NMR

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4.4: Synthesis and Characterization of a Metal Hydride Complex

Learning Objectives

- Synthesize a cobalt hydride complex with phosphite ligands.
- Isolate and purify the product, recording key physical properties.
- Analyze ^1H and ^{31}P NMR data to propose a plausible molecular formula and structure for the product.
- Apply the 18-electron rule and assign the oxidation state and electron count.
- Explain the role of NaBH_4 in the formation of the product.

Introduction

A hydride is a chemical species in which hydrogen is formally present as H^- . Metal hydrides are an important class of compounds, with the hydrogen atom acting as an anionic ligand. The $\text{M}-\text{H}$ bond appears in a variety of contexts, including as a component of reaction intermediates, and it plays a key role in chemical transformations such as oxidative addition and reductive elimination.

Metal hydrides are also crucial intermediates in homogeneous catalysis, including hydrogenation, hydroformylation, and $\text{C}-\text{H}$ activation. They have significant synthetic applications, such as enabling hydride transfer reactions (e.g., hydrometallation, hydrosilylation), and they promote selective reductions of organic substrates under relatively mild conditions.

Additionally, metal hydrides serve as models for hydrogen activation in both synthetic and biological contexts. They mimic reactive intermediates found in hydrogenase and nitrogenase enzymes and are valuable tools for studying reaction mechanisms—particularly those involving redox and proton-coupled electron transfer (PCET) processes.

In this experiment, you will synthesize and characterize a transition-metal hydride complex containing cobalt. After preparing the compound, you will analyze it using several spectroscopic techniques, including IR, ^1H NMR, ^{13}C NMR, and $^1\text{H}\{^{31}\text{P}\}$ NMR, to deduce its structure.

Experimental Method

Safety Precautions

- Cobalt nitrate hydrate is an oxidizing and toxic heavy-metal compound; avoid skin and eye contact and wear gloves and goggles when handling it.
- Sodium borohydride reacts with water and alcohols; add it slowly to ethanol. It releases flammable H_2 gas, so keep it away from ignition sources.
- Triphenyl phosphite ($\text{P}(\text{OPh})_3$) is an irritant; avoid inhalation and skin contact and handle it in a fume hood.
- Dichloromethane (DCM) is volatile and a possible carcinogen; use it in a well-ventilated fume hood and avoid inhalation.
- Ethanol is highly flammable; keep it away from flames and sparks.
- Wear a lab coat, gloves, and splash goggles at all times.
- Perform all steps involving DCM, ethanol, or phosphites in a fume hood.
- Collect DCM waste and cobalt-containing waste in designated waste containers.

Materials

Chemical Name	Formula	CAS Number	Molar Mass (g/mol)	Density (g/mL)
Cobalt(II) nitrate hexahydrate	$\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	10026-22-9	291.03	
Sodium borohydride	NaBH_4	16940-66-2	37.83	
Triphenyl phosphite	$\text{P}(\text{OC}_6\text{H}_5)_3$	101-02-0	310.28	1.18
Dichloromethane	CH_2Cl_2	75-09-2	84.93	1.33
Ethanol	$\text{C}_2\text{H}_5\text{OH}$	64-17-5	46.07	0.789
Water	H_2O	7732-18-5	18.02	0.998
Methanol	CH_3OH	67-56-1	32.04	0.792

Procedure

1. Using a 25 mL Erlenmeyer flask, dissolve 0.5 g of sodium borohydride in 10 mL of ethanol.
2. In a 125 mL Erlenmeyer flask, dissolve 1.5 g of cobalt(II) nitrate hydrate and 8.0 g of triphenylphosphite in 30 mL of ethanol. Add a magnetic stir bar and stir continuously.
3. Using a Pasteur pipet, add the previously prepared sodium borohydride solution dropwise to the stirring solution containing cobalt(II) nitrate hydrate and triphenylphosphite.
4. After 15 minutes, collect the precipitate by filtration using a Büchner funnel. Wash the product sequentially with ethanol, water, and methanol. Allow the product to dry.
5. Recrystallize the product by dissolving the solid in a minimum amount of dichloromethane (DCM). Add DCM to the solid in small (~5 mL or less) portions until the primary solid has dissolved. This may require about 30 mL of DCM. Note that some insoluble material that is not the product may remain.
6. Filter the solution through fluted filter paper using a wide-mouth funnel, or through a small plug of cotton in a standard funnel, to obtain a colored but clear solution.
7. Once a clear solution is obtained, recrystallize the product by slowly adding ethanol until a precipitate forms.
8. Record the yield and melting point.
9. Record the IR spectrum, ^1H , ^{13}C , and $^{31}\text{P}\{^1\text{H}\}$ NMR.

Data Collection and Analysis

1. Describe the color of the product
2. Calculate the % yield. The calculation must be in your laboratory notebook.
3. Analyze and interpret the NMR data. Discuss what the observed peaks or peak patterns reveal in each spectrum and use these observations to propose a structure for the compound. You may reference literature sources to support your interpretation.

Questions and Further Thoughts

1. Based on your data and proposed structure, determine the oxidation state of the cobalt.
2. Does the complex satisfy the 18-electron rule?
3. What is the geometry around the cobalt center? Justify your answer based on electron count and ligand type.
4. What observations during the reaction might have indicated that hydrogen gas was being produced? What might be the cause and how do you think this would affect the yield of your product? What steps could be taken to minimize or eliminate this reaction?

Notes

Further Readings

1. Platt, A. W. G. Synthesis and Characterisation of a Metal Hydride Complex. In *Inorganic Experiments*; Woollins, J. D., Ed.; Wiley-VCH: Weinheim, Germany, 2003; pp 63–64
2. Crane, J. D.; Young, N. Hydridotetrakis(Triphenylphosphito)Cobalt(I). *Acta Crystallogr E Struct Rep Online* **2004**, 60 (4), m487–m488. <https://doi.org/10.1107/S1600536804007093>.

Techniques: recrystallization, filtration, ^1H NMR, ^{13}C NMR, ^{31}P NMR

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4.5: Ni(II) Coordination Complexes

Learning Objectives

- Identify possible geometric isomers of nickel(II) coordination complexes based on coordination number and ligand arrangement.
- Use infrared spectroscopy data to distinguish between possible structural isomers of coordination compounds.
- Collect and interpret magnetic susceptibility data to determine whether a nickel complex is diamagnetic or paramagnetic and relate this to its geometry.
- Apply ligand field theory concepts to explain differences in magnetic behavior and geometry among nickel(II) complexes.
- Correlate experimental data with molecular structure to determine which structural model best fits the observations.

Introduction

Coordination compounds can exist in a variety of geometric isomers. Geometric isomers are a type of stereoisomer in which the same atoms are connected in the same order but are oriented differently in space around the metal center. Cis–trans isomerism is one of the most common forms of geometric isomerism. In inorganic chemistry, geometric isomerism commonly occurs in square-planar and octahedral complexes.

In square-planar complexes with the general formula MA_2B_2 , cis–trans isomers can exist where the cis form has two identical ligands adjacent (90° apart), while in the trans form the identical ligands are opposite (180° apart). Cis–trans isomerism can also occur in octahedral complexes. For example, in complexes with the general formula MA_4B_2 , the B ligands can either be cis (adjacent and 90° apart) or trans (opposite and 180° apart).

Facial–meridional (fac–mer) isomerism can also occur in some octahedral complexes. Complexes with the general formula MA_3B_3 have a facial isomer in which the three identical ligands occupy one triangular face of the octahedron (all 90° to each other) and a meridional isomer in which the three identical ligands lie in a plane passing through the metal center, with two trans and one cis to the others. Other forms of geometric isomerism also exist.

Geometric isomers can have very different properties, including reactivity, spectroscopic signatures, color, stability, and biological activity. For example, the cis and trans forms of $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$ have dramatically different biological properties.

Because geometric isomers have different properties, it is possible to differentiate between the different forms using experimental data. In this experiment, you will analyze infrared spectroscopy and magnetic susceptibility data collected for three different nickel(II) coordination complexes, all of which can exist as geometric isomers. For two of the complexes, you will determine which of three possible geometric isomers is most consistent with your data.

Experimental Method

Safety Precautions

- Nickel salts and phosphines are toxic—handle them with care.
- Phosphines can be irritants; avoid skin contact and inhalation.
- All reactions must be carried out in a fume hood.
- Dispose of all waste in the appropriate designated waste container.
- Nickel(II) chloride ($\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$) is toxic and a skin sensitizer. Avoid inhalation and skin contact, and wear gloves, safety goggles, and a lab coat when handling nickel salts. Dispose of all nickel-containing waste in the designated heavy-metal waste container.
- Ethylenediamine is corrosive and irritating to the skin, eyes, and respiratory system. Handle it in a fume hood and avoid inhalation or direct contact.
- Ethanol and diethyl ether are highly flammable and volatile. Handle them in a fume hood and keep them away from flames, sparks, and other ignition sources.
- Add ethylenediamine slowly while the solution is cooled to control heat generation and minimize splashing.

Materials

Chemical Name	Formula	CAS Number	Molar Mass (g/mol)	Density (g/mL)

Nickel(II) chloride hexahydrate	$\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$	7791-20-0	237.69	— (solid)
Nickel(II) nitrate hexahydrate	$\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	13478-00-7	290.81	— (solid)
Sodium thiocyanate	NaSCN	540-72-7	81.07	— (solid)
Ethylenediamine	$\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$	107-15-3	60.10	0.899
Triphenylphosphine	$\text{P}(\text{C}_6\text{H}_5)_3$	603-35-0	262.29	— (solid)
Ethanol (absolute)	$\text{C}_2\text{H}_5\text{OH}$	64-17-5	46.07	0.789
2-Propanol (isopropanol)	$\text{C}_3\text{H}_8\text{O}$	67-63-0	60.10	0.785
Diethyl ether	$(\text{C}_2\text{H}_5)_2\text{O}$	60-29-7	74.12	0.713
Distilled (or deionized) water	H_2O	7732-18-5	18.02	1.000

Procedure

Synthesis of $[\text{Ni}(\text{en})_3]\text{Cl}_2 \cdot \text{H}_2\text{O}$

1. In a small Erlenmeyer flask (≈ 25 mL), dissolve 1.5 g of $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ in 2 mL of distilled water.
2. Stir until the solid has completely dissolved.
3. Cool the solution in an ice bath.
4. While maintaining cooling, slowly add 1.4 mL of ethylenediamine dropwise.
5. After the addition is complete, stir the mixture for 5 minutes.
6. Add 10 mL of cold ethanol to induce crystallization of the product.
7. While continuing to cool, stir the mixture for an additional 10 minutes.
8. Isolate the product by suction filtration, then wash with cold absolute ethanol (2×5 mL) followed by diethyl ether (3×5 mL).
9. Dry the product in place, then transfer it to a vial and store it in a desiccator or drybox until the next class period.

Synthesis of $\text{NiCl}_2(\text{PPh}_3)_2$

1. In a 100 mL round-bottom flask, add 2.8 g of triphenylphosphine and 30 mL of dry 2-propanol.
2. Add two boiling stones, attach a reflux condenser, and bring the mixture to reflux.
3. In a 50 mL Erlenmeyer flask, dissolve 1.2 g of nickel(II) chloride hexahydrate in 15 mL of ethanol, then warm the solution to 40°C .
4. Carefully pour the warm nickel solution down the condenser into the refluxing triphenylphosphine solution.
5. Observe the formation of a precipitate.
6. Allow the mixture to cool to room temperature.
7. Filter the product using a Büchner funnel fitted with a sintered glass fiber filter.
8. Wash the precipitate on the filter with 15 mL of cold ethanol.
9. Allow the product to dry on the funnel.
10. Transfer the solid to a vial and store it in a desiccator until the next class period.

Synthesis of $\text{NiCl}_2(\text{NCS})_2$

1. In a 100 mL round bottom flask along with two antidumping stones dissolve 1.5 g $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in 25 mL ethanol.
2. Add 0.8 g finely powdered sodium thiocyanate.
3. Fit the flask with a condenser and reflux for 20 minutes.
4. In separate 100 mL round-bottom flask, add 2.8 g of triphenylphosphine and 30 mL of dry 2-propanol.
5. Add two boiling stones, attach a reflux condenser, and bring the mixture to reflux.
6. After 20 mins cool the thiocyanate solution in an ice bath.
7. Scratch the bottom of the flask with a glass rod to induce precipitation of sodium nitrate and unreacted sodium thiocyanate.
8. Filter off and set aside the solids and retain the filtrate.

9. Transfer the filtrate into an Erlenmeyer flask along with 1-2 boiling stones.
10. Heat the solution then carefully pour it down the reflux condenser into the refluxing phosphine solution prepared in #4 above.
11. After the addition, allow the reaction mixture to cool to room temperature.
12. Filter off the product using a Büchner funnel.
13. Wash the product with cold ethanol (~15 mL) followed by cold dry ether (~15 mL)
14. Allow the product to dry in place.

Data Collection and Analysis

1. Calculate the percent yield for each nickel complex synthesized.
2. Compare the observed colors of the complexes and relate them to differences in ligand field strength and geometry.
3. Assign key IR peaks for each complex, especially N–H stretches for ethylenediamine, P–C/PPh₃-related bands, and SCN stretching bands.
4. Use IR data to determine whether thiocyanate is N-bound or S-bound in the nickel thiocyanate complex.
5. Determine the number of unpaired electrons from magnetic susceptibility data.
6. Use magnetic data to distinguish between diamagnetic square-planar Ni(II) complexes and paramagnetic tetrahedral or octahedral Ni(II) complexes.
7. Compare experimental magnetic moments to spin-only values for Ni(II).
8. Use ligand field theory to explain why different ligands produce different geometries around Ni(II).
9. Compare the experimental results for [Ni(en)₃]Cl₂, NiCl₂(PPh₃)₂, and Ni(NCS)₂(PPh₃)₂ or related thiocyanate complexes.
10. Discuss possible sources of error, including incomplete drying, product loss during filtration, incomplete reaction, and contamination with starting material.

Questions and Further Thoughts

1. Why do different ligands cause Ni(II) to adopt different coordination geometries?
2. The complex [Ni(en)₃]²⁺ is generally more stable than the analogous ammonia complex. Explain this difference using the chelate effect.
3. How can magnetic susceptibility measurements distinguish between square-planar and tetrahedral Ni(II) complexes?
4. How can IR spectroscopy be used to determine whether SCN[−] is coordinated through nitrogen or sulfur?
5. Compare the bonding properties of ethylenediamine, triphenylphosphine, and thiocyanate, and explain how they influence the properties of the nickel center.

Notes

Further Readings

1. Batsanov, A. S.; Howard, J. A. K. *Trans* -Dichlorobis(Triphenylphosphine)Nickel(II) Bis(Dichloromethane) Solvate: Redetermination at 120 K. *Acta Crystallogr E Struct Rep Online* **2001**, 57 (7), m308–m309. <https://doi.org/10.1107/S1600536801008741>.
2. Brammer, L.; Stevens, E. D. Structure of Dichlorobis(Triphenylphosphine)Nickel(II). *Acta Crystallogr C Cryst Struct Commun* **1989**, 45 (3), 400–403. <https://doi.org/10.1107/S0108270188011692>.
3. Corain, B.; Longato, B.; Angeletti, R.; Valle, G. *Trans*-[Dichlorobis(Triphenylphosphine)Nickel(II)]·(C₂H₄Cl₂)₂: A Clathrate of the Allogon of Venanzi's Tetrahedral Complex. *Inorganica Chimica Acta* **1985**, 104 (1), 15–18. [https://doi.org/10.1016/S0020-1693\(00\)83780-9](https://doi.org/10.1016/S0020-1693(00)83780-9).
4. Garton, G.; Henn, D. E.; Powell, H. M.; Venanzi, L. M. 682. Tetrahedral Nickel(II) Complexes and the Factors Determining Their Formation. Part V. The Tetrahedral Co-Ordination of Nickel in Dichlorobistriphenylphosphinenickel. *J. Chem. Soc.* **1963**, 3625. <https://doi.org/10.1039/jr9630003625>.

5. Figgis, B. N.; Lewis, J. The Magnetic Properties of Transition Metal Complexes. In *Progress in Inorganic Chemistry*; Cotton, F. A., Ed.; Wiley, 1964; Vol. 6, pp 37–239. <https://doi.org/10.1002/9780470166079.ch2>.

Techniques: reflux, filtration, magnetic susceptibility, IR spectroscopy

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4.6: Preparation of (benzene)tricarbonylchromium(0)

Learning Objectives

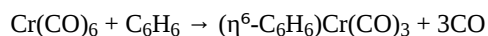
- Perform the synthesis and isolation of an arene–metal tricarbonyl complex.
- Apply safe operating procedures for high-pressure chemical reactions.
- Interpret infrared spectroscopic data to characterize the compound.
- Describe the bonding interaction between chromium and the coordinated η^6 -arene ligand and explain how π -backbonding is reflected in the spectroscopic data.
- Perform a recrystallization.

Introduction

Metal carbonyl compounds are among the most important compounds in organometallic chemistry. They serve as important precursors for the synthesis of many organometallic complexes and are involved in several major catalytic processes in industry. Studying these complexes helps illustrate fundamental principles such as metal–ligand bonding, electron counting, and catalytic reactivity.

Carbon monoxide binds to a metal through σ -donation from the carbon lone pair to an empty orbital on the metal while simultaneously accepting electron density from filled metal d-orbitals into the π^* antibonding orbitals of CO. This interaction is known as π -backbonding. The extent of π -backbonding, or simply backbonding, influences the strength of the C–O bond and is reflected in the C–O stretching frequencies observed using infrared spectroscopy. In general, stronger metal-to-CO backbonding weakens the C–O bond, leading to lower C–O stretching frequencies. Electron density at the metal available for backbonding is determined and influenced by all of the ancillary ligands bound to the metal. Electron-donating ligands provide more electron density to the metal while electron-withdrawing ligands reduce it.

Arene-metal complexes are a class of organometallic compounds in which aromatic rings bind to the metal through their π -electron system. Chromium(0) hexacarbonyl reacts with benzene to form benzene tricarbonylchromium(0) according to the following reaction:



For this reaction, benzene serves as both the solvent and the reactant. To produce the product, a reaction temperature of 160–175 °C is required. However, the boiling point of benzene is only 80 °C. If we attempted to carry out the reaction in standard laboratory glassware at these temperatures, the benzene would boil away before the reaction could occur. To prevent this, the reaction is carried out in a high-pressure reactor (autoclave), or “bomb.” Because the reactor is sealed, the benzene cannot escape, and its vapor pressure increases so that it can reflux in the 160–175 °C range.

In this experiment, you will utilize a high-pressure reactor to synthesize (benzene)tricarbonylchromium(0).

Experimental Method

Safety Precautions

- Chromium hexacarbonyl ($\text{Cr}(\text{CO})_6$) is highly toxic and may release carbon monoxide (CO), a colorless, odorless, highly toxic gas, if heated or decomposed. Avoid inhalation and skin contact, and conduct all operations involving $\text{Cr}(\text{CO})_6$ in a fume hood while wearing gloves, safety goggles, and a lab coat.
- Benzene is a known human carcinogen. Avoid inhalation and skin contact. Keep containers closed and handle only in a fume hood using appropriate PPE.
- THF is highly flammable, and older samples may contain explosive peroxides. Keep away from ignition sources and use fresh or peroxide-tested THF.
- The sealed reactor may generate pressure during heating. Inspect the vessel and seals before use and do not exceed recommended temperature or pressure limits.
- The reactor and liner may become very hot during the reaction. Allow them to cool and use heat-resistant gloves when handling.
- Open the reactor slowly and only in a fume hood to release pressure safely. Stand to the side of the vessel while opening.
- Dispose of benzene solutions and chromium-containing waste in designated hazardous-waste containers. Do not pour down the drain.

Materials

Chemical Name	Formula	CAS Number	Molar Mass (g/mol)	Density (g/mL)
Chromium hexacarbonyl	$\text{Cr}(\text{CO})_6$	13007-92-6	220.06	— (solid)
Benzene	C_6H_6	71-43-2	78.11	0.879
Tetrahydrofuran (THF)	$\text{C}_4\text{H}_8\text{O}$	109-99-9	72.11	0.889

Caution

Benzene is a known human carcinogen. Avoid inhalation and skin contact. Keep containers closed and handle only in a fume hood using appropriate PPE including nitrile gloves and a lab coat.

Procedure

1. Weigh out 2.0 g of $\text{Cr}(\text{CO})_6$ and place it in the glass liner of the reactor.
2. Add 10 mL of benzene to the stainless-steel reactor.
3. Carefully place the glass liner containing $\text{Cr}(\text{CO})_6$ into the reactor.
4. Add 30 mL of benzene to the glass liner.
5. Add 2 mL of THF to the glass liner to serve as a catalyst.
6. Assemble the reactor according to the instructor's or TA's instructions.
7. Inspect the reactor to ensure that it is properly assembled and that all fittings are securely tightened before heating.
8. Allow the reaction to proceed in the sealed reactor at 160–175 °C for 24–48 hours.
9. Once the reaction is complete, allow the reactor to cool to room temperature.
10. Slowly release the pressure and disassemble the reactor.
11. While wearing nitrile gloves, transfer the reaction mixture to a round-bottom flask and remove the solvent using a rotary evaporator.
12. Obtain an IR spectrum of the crude product.
13. Purify the product by dissolving it in a minimum volume of boiling hexane. Filter the hot solution and allow it to cool in an ice bath to crystallize the product.
14. Isolate the solid by vacuum filtration using a Büchner funnel.
15. Dry the product in a desiccator.
16. Determine the mass of the purified product and calculate the percent yield.
17. Obtain an IR spectrum of the purified product.
18. Obtain an IR spectrum of the starting material.

Data Collection and Analysis

1. Percent Yield

- a. Calculate the theoretical yield of $(\eta^6\text{-C}_6\text{H}_6)\text{Cr}(\text{CO})_3$ from the amount of $\text{Cr}(\text{CO})_6$ used.
- b. Determine the percent yield of the isolated product.
- c. Identify the limiting reagent and comment on factors that may have reduced the yield.

2. Infrared Spectroscopic Analysis

- a. Assign the carbonyl stretching bands (ν_{CO}) observed in the IR spectrum.
- b. Compare the experimental ν_{CO} values with literature values.
- c. Explain how π -backbonding between chromium and the CO ligands influences the observed stretching frequencies.

3. Electron Counting and Bonding

- a. Determine the oxidation state of chromium in $(\eta^6\text{-C}_6\text{H}_6)\text{Cr}(\text{CO})_3$.
- b. Perform an 18-electron count using the neutral ligand method (or ionic method, if discussed in class).
- c. Explain how η^6 -coordination of benzene contributes to the electron count and the stability of the complex.

Questions and Further Thoughts

1. Why is a high-pressure reactor required for the synthesis of $(\eta^6\text{-C}_6\text{H}_6)\text{Cr}(\text{CO})_3$?
2. Explain how the η^6 -coordination of benzene differs from the way benzene interacts in ordinary organic reactions.
3. Why do the C–O stretching frequencies in $(\eta^6\text{-C}_6\text{H}_6)\text{Cr}(\text{CO})_3$ differ from those of free carbon monoxide?
4. Using the 18-electron rule, explain why $(\eta^6\text{-C}_6\text{H}_6)\text{Cr}(\text{CO})_3$ is particularly stable.
5. Predict how replacing benzene with a more electron-donating arene (e.g., anisole or mesitylene) would affect the CO stretching frequencies in the IR spectrum. Explain your reasoning.
6. Chromium hexacarbonyl contains six CO ligands, whereas the product contains only three. What drives the substitution of three CO ligands by benzene under the reaction conditions?
7. Suggest at least three factors that could lower the isolated yield of $(\eta^6\text{-C}_6\text{H}_6)\text{Cr}(\text{CO})_3$.
8. Why should the reactor be opened slowly and only after it has cooled to room temperature?
9. If the product contained unreacted $\text{Cr}(\text{CO})_6$, how might you detect this?

Notes

Further Readings

1. Angelici, R. J. *Synthesis and Technique in Inorganic Chemistry*, 3rd ed.; University Science Books: Sausalito, CA, 1999; pp. 129–146.
2. Fisher, R.D. *Chem. Ber.* 1960, 93, 165-175.

Techniques: high pressure synthesis, recrystallization, sublimation, NMR, and IR

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4.7: Synthesis of a High Temperature Superconductor

Learning Objectives

- Describe the structure and composition of $\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$ and explain the role of oxygen stoichiometry in determining its superconducting properties.
- Calculate appropriate reagent quantities required to obtain the correct stoichiometric ratio of Y, Ba, and Cu in $\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$.
- Carry out a solid-state ceramic synthesis using high-temperature reactions between metal oxides.
- Perform high-temperature solid-state reactions safely, including grinding, mixing, calcination, and sintering steps.
- Obtain a powder X-ray diffraction (PXRD) spectrum and interpret it by comparison with reference patterns or crystallographic databases to assess the identity and phase purity of the sample.
- Recognize and interpret qualitative evidence of superconductivity, including magnetic levitation arising from the Meissner effect and flux pinning.
- Analyze potential sources of error in solid-state synthesis, such as incomplete mixing, insufficient heating, or incorrect stoichiometry.

Introduction

$\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$ (YBCO), where x ranges from 0 to about 0.65, is a well-known example of a high-temperature superconducting material. One way to prepare YBCO is by reacting oxides or carbonates of barium, yttrium, and copper in a muffle furnace. By superconducting, it is meant that the material loses its resistance to electrical current and expels magnetic fields. Many conductors lose some electrical resistance as temperature decreases. Superconductors are unique because their resistance decreases abruptly below a temperature known as the critical temperature.

The superconducting properties of YBCO are sensitive to the stoichiometry of oxygen. High-quality YBCO can be prepared by carefully controlling the oxygen stoichiometry, but material of suitable quality can also be conveniently prepared in air using a muffle furnace.

YBCO crystallizes in a layered defect perovskite structure. An ideal perovskite has the general formula ABO_3 . In $\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$, some oxygen sites are vacant, giving the composition $\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$.

In this experiment, you will prepare YBCO in a muffle furnace and test it for superconducting properties by cooling the sample with liquid nitrogen and attempting to float a magnet above the material, illustrating the Meissner effect.

Experimental Method

Safety Precautions

- Copper, barium, and yttrium oxides may be harmful if inhaled or ingested. Weigh and mix powders in a fume hood or well-ventilated area. Avoid skin contact, wear gloves when handling, and wash hands after completing the experiment.
- Use caution when operating the muffle furnace. The furnace and ceramic crucibles become extremely hot and can cause severe burns.
- Use heat-resistant gloves or crucible tongs when placing materials in or removing materials from the furnace and allow hot crucibles or samples to cool on a heat-resistant surface before handling.
- Liquid nitrogen can cause severe cold burns and frostbite. Handle it with insulated gloves and safety goggles.
- Use liquid nitrogen only in well-ventilated areas.
- Do not touch objects that have been cooled with liquid nitrogen with bare hands.

Materials

Chemical Name	Formula	CAS Number	Molar Mass (g/mol)	Density (g/mL)
Yttrium(III) oxide	Y_2O_3	1314-36-9	225.81	— (solid)
Barium peroxide	BaO_2	1304-29-6	169.33	— (solid)
Copper(II) oxide	CuO	1317-38-0	79.55	— (solid)
Liquid nitrogen	N_2	7727-37-9	28.01	0.808*

*Density of liquid nitrogen at its normal boiling point ($-196\text{ }^{\circ}\text{C}$, 1 atm).

Procedure

1. The final YBCO product contains yttrium, barium, and copper in a $1\text{Y}:2\text{Ba}:3\text{Cu}$ ratio. Calculate the masses of BaO_2 and CuO required to react with 1.20 g of Y_2O_3 . Since this reaction is conducted in air, ignore the amount of oxygen.
2. Have the calculated masses approved by your instructor before proceeding.
3. Accurately weigh out 1.20 g of yttrium(III) oxide.
4. Weigh out the stoichiometrically equivalent amounts of barium peroxide and copper(II) oxide.
5. Combine all three oxides in a mortar.
6. Mix and grind the reactants with a pestle until the mixture has a uniform color and consistency.
7. If available, obtain a powder X-ray diffraction pattern of this mixture.
8. Transfer the mixture into a pellet press.
9. Press a pellet by applying a constant pressure of 15,000 lb for 15 minutes.
10. After 15 minutes, carefully remove the pellet from the press. The pellet is extremely fragile and may crumble. If it does, repeat Steps 6–9.
11. Carefully position the pellet on top of a narrow crucible and place it into a muffle furnace.
12. Program the furnace to heat from $25\text{ }^{\circ}\text{C}$ to $930\text{ }^{\circ}\text{C}$ at a rate of $1.89\text{ }^{\circ}\text{C}/\text{min}$. Hold at $930\text{ }^{\circ}\text{C}$ for 12 hours. Lower the temperature to $500\text{ }^{\circ}\text{C}$ at a rate of $1.89\text{ }^{\circ}\text{C}/\text{min}$. Hold at $500\text{ }^{\circ}\text{C}$ for 12 hours. Lower the temperature to $25\text{ }^{\circ}\text{C}$ at a rate of $1.89\text{ }^{\circ}\text{C}/\text{min}$. The complete furnace program requires approximately 40 hours.
13. Remove the pellet from the furnace.
14. Describe the appearance of the pellet. The finished pellet should be dark gray to black. A green material indicates the presence of Y_2BaCuO_5 , a non-superconducting phase.
15. Place the pellet on top of an upside-down foam cup or in a Petri dish.
16. Place a small magnet on the pellet. Observe.
17. Remove the magnet.
18. Pour liquid nitrogen onto the pellet to cool it.
19. Place the magnet back on the pellet. Observe. The magnet should levitate above the superconducting pellet. This is called the Meissner effect.
20. Use a pair of forceps to touch the corner of the magnet. It should spin above the pellet.
21. Return the pellet to a mortar and crush it with a pestle.
22. Obtain a powder X-ray diffraction pattern of the superconducting material.

Data Analysis

1. Determine the theoretical yield of $\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$ based on the limiting reagent.
2. Obtain powder X-ray diffraction (PXRD) patterns of the sample before and after firing. Compare the diffraction patterns and discuss the evidence for formation of the $\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$ phase.
3. Compare the observed product color and texture before and after firing/grinding.
4. Analyze the magnetic response of the sample before and after cooling with liquid nitrogen.
5. Describe the evidence for superconductivity based on the Meissner effect or magnetic levitation.
6. Discuss how incomplete reaction, poor pellet formation, oxygen deficiency, or insufficient firing could affect the observed PXRD pattern and superconducting behavior.

Questions and Further Thoughts

1. Why is oxygen essential for the superconductivity of $\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$?
2. What is the Meissner effect, and how does it differ from ordinary diamagnetism?
3. Compare the powder X-ray diffraction patterns obtained before and after firing. What evidence indicates that $\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$ was formed?
4. Why is the firing schedule (high-temperature sintering followed by slow cooling) important for obtaining a superconducting sample?
5. Suggest three factors that could prevent the sample from exhibiting superconductivity, even if the synthesis was carried out correctly.

Notes

Further Readings

She, J.-L.; Liu, R.-S. A Simplified Synthetic Experiment of $\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$ Superconductor for First-Year Chemistry Laboratory. *J. Chem. Educ.* **2008**, 85 (6), 825. <https://doi.org/10.1021/ed085p825>.

Cogdell, C. D.; Wayment, D. G.; Casadonte, D. J.; Kubat-Martin, K. A. A Convenient, One-Step Synthesis of $\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$ Superconductors: An Undergraduate Inorganic/Materials Laboratory Experiment. *J. Chem. Educ.* **1995**, 72 (9), 840. <https://doi.org/10.1021/ed072p840>.

Djurovich, P. I.; Watts, R. J. A Simple and Reliable Chemical Preparation of $\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$ Superconductors: An Experiment in High Temperature Superconductivity for an Advanced Undergraduate Laboratory. *J. Chem. Educ.* **1993**, 70 (6), 497. <https://doi.org/10.1021/ed070p497>.

Green, D. B.; Douphner, D.; Hutchinson, B. A Simple, Inexpensive Device for Measuring the Critical Temperature of a High-Temperature Superconductor. *J. Chem. Educ.* **1992**, 69 (4), 343. <https://doi.org/10.1021/ed069p343>.

Techniques: solid-state “heat and beat” high-temperature synthesis, measurement of critical temperature, T_c , of a superconductor, powder X-ray diffraction

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4.8: Cobalt Complex of Dioxygen

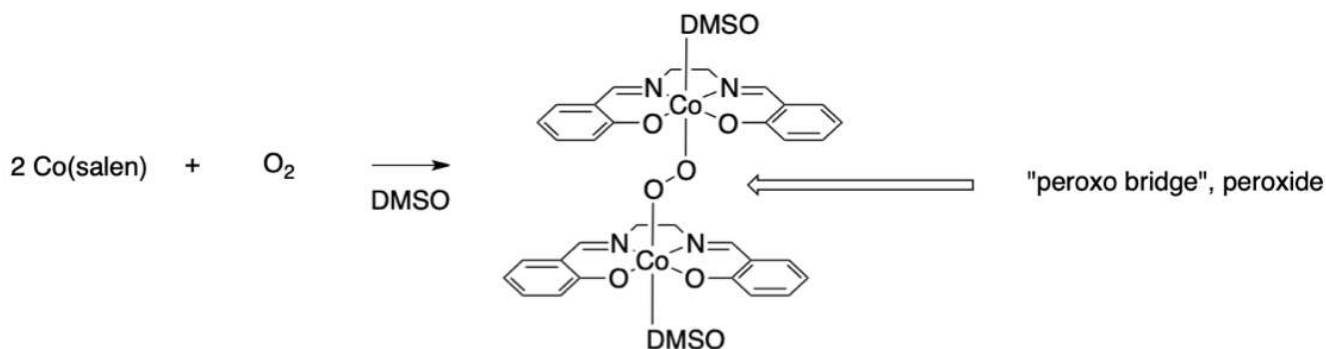
Learning Objectives

- Synthesize the Schiff-base ligand salen through the condensation reaction of salicylaldehyde and ethylenediamine and identify the formation of imine functional groups characteristic of Schiff bases.
- Prepare a cobalt coordination complex by metallating the tetradentate salen ligand and recognize how multidentate ligands bind to metal centers through multiple donor atoms.
- Describe the coordination geometry of cobalt in salen complexes and explain how chelation by a tetradentate ligand stabilizes the metal center.
- Observe and explain reversible oxygen binding in a cobalt salen complex, including the associated redox process in which Co(II) is oxidized to Co(III) and O₂ is reduced to a peroxide species.

Introduction

Cobalt Schiff-base complexes were among the first synthetic model oxygen carriers studied during the 1960s–1970s as models for biological oxygen transport in hemoglobin and myoglobin. Although they are not perfect analogs, these systems are similar in that both reversibly bind O₂ and the metal center is oxidized upon oxygenation.

One example of Schiff-base complexes capable of reversibly binding oxygen is the class of cobalt salen complexes (salen = bis(salicylaldehyde)ethylenediamine). These complexes have been historically important in coordination and bioinorganic chemistry. The cobalt(II) salen complex binds oxygen according to the following equilibrium:



In this process, the cobalt(II) centers are oxidized to Co(III), while O₂ is reduced to peroxide. The peroxide ligand typically bridges two cobalt centers to form a μ -peroxo dicobalt(III) complex.

The position of the equilibrium depends on O₂ pressure, solvent, temperature, and the presence of axial ligands.

In this experiment, you will prepare a cobalt salen complex and observe its ability to reversibly bind oxygen.

Experimental Method

Safety Precautions

- Ethanol and diethyl ether are highly flammable. Keep away from flames and ignition sources and heat ethanol only on a hotplate in a well-ventilated area.
- Salicylaldehyde is an irritant with a strong odor. Avoid inhalation and skin contact and handle it in a fume hood while wearing gloves and safety goggles.
- Ethylenediamine is corrosive and strongly irritating to the skin, eyes, and respiratory system. Avoid contact and inhalation and handle it only in a fume hood with appropriate PPE.
- Cobalt salts are toxic heavy-metal compounds. Avoid skin contact and inhalation and dispose of cobalt-containing waste in designated heavy-metal waste containers.
- DMSO readily penetrates the skin and can carry dissolved chemicals with it. Avoid skin contact and wear gloves when handling DMSO solutions.

- Dichloromethane (CH_2Cl_2) is volatile and potentially harmful if inhaled and is classified as a suspected carcinogen. Handle it only in a fume hood and avoid breathing vapors.

Materials

Chemical Name	Formula	CAS Number	Molar Mass (g/mol)	Density (g/mL)
Salicylaldehyde (2-hydroxybenzaldehyde)	$\text{C}_7\text{H}_6\text{O}_2$	90-02-8	122.12	1.14
Ethylenediamine	$\text{C}_2\text{H}_8\text{N}_2$	107-15-3	60.10	0.90
Ethanol	$\text{C}_2\text{H}_5\text{OH}$	64-17-5	46.07	0.789
Cobalt(II) acetate tetrahydrate	$\text{Co}(\text{CH}_3\text{COO})_2 \cdot 4\text{H}_2\text{O}$	6147-53-1	249.08	— (solid)
Water (deionized)	H_2O	7732-18-5	18.02	1.00
Dimethyl sulfoxide (DMSO)	$(\text{CH}_3)_2\text{SO}$	67-68-5	78.13	1.10
Diethyl ether	$(\text{C}_2\text{H}_5)_2\text{O}$	60-29-7	74.12	0.713
Dichloromethane (Methylene chloride)	CH_2Cl_2	75-09-2	84.93	1.33

Procedure

Preparation of “salenH₂”

- Obtain 10.0 mL of ethanol in a small conical flask (e.g. 25 mL).
- Carefully stir and boil the ethanol on a hotplate.
- Add 0.90 mL of salicylaldehyde to the ethanol.
- Add 250 μL of ethylenediamine dissolved in 1 mL of ethanol to the flask.
- Stir the flask for 5 mins then place it into an ice bath.
- Collect the product by suction filtration.
- Carefully wash the product with 2 mL of cold ethanol.
- Obtain the mass of the product. You will use some of your product in the next section. Be sure to retain enough to obtain ATR-IR and ^1H NMR spectra.

Preparation of a Schiff base complex “Co(salen)”

This must be done under an inert atmosphere.

- Set up an apparatus for reflux in an inert atmosphere.
- Place 230 mg of salenH₂ in the flask.
- Add 12 mL of ethanol to the flask.
- Flush the system with argon and heat to about 70 – 80 °C. (important)
- Add a solution of $\text{Co}(\text{acetate})_2 \cdot 4\text{H}_2\text{O}$ (200 mg in 1.5 mL water) to the flask.
- Continue to heat and stir for 1 hr. Be sure to record your observations!!
- After 1 hr, place flask in an ice bath.
- Isolate the product by suction filtration.
- Carefully wash the product with a few drops of cold ethanol.
- Weight the dried product.

11. Obtain ATR-IR and magnetic susceptibility.

Oxygenation of “Co(salen)”

Caution

DMSO is readily absorbed by the skin and will carry other compounds through the skin with it. Be Careful. If exposed, wash thoroughly with water

1. Stir about 0.1 g of Co(salen) in 20 mL of DMSO for 1 hour
2. Collect the product by suction filtration. (record your observations!)
3. Wash the product with diethyl ether.
4. Take the product into a small beaker and carefully add CH_2Cl_2 to all of it.
5. Record your observations! Be prepared to explain them in your discussion.

Data Collection and Analysis

1. Record all observations during the synthesis of H_2salen , Co(salen), and the oxygenation reaction.
2. Calculate the theoretical and percent yields of H_2salen and Co(salen).
3. Obtain and interpret the ATR-IR spectra of H_2salen and Co(salen). Compare the spectra and identify evidence of metal coordination.
4. Obtain and interpret the ^1H NMR spectrum of H_2salen to confirm formation of the Schiff base ligand.
5. Measure the magnetic susceptibility of Co(salen), calculate the effective magnetic moment, estimate the number of unpaired electrons, and relate your observations from the oxygenation experiment to the reversible binding of oxygen.

Questions and Further Thoughts

1. Explain why an inert atmosphere is required during the synthesis of Co(salen).
2. Compare the ATR-IR spectra of H_2salen and Co(salen). Which spectral changes provide evidence that coordination to cobalt has occurred?
3. Using your magnetic susceptibility data, determine the number of unpaired electrons in Co(salen). Does your result agree with the expected electronic structure?
4. Draw the structure of Co(salen) and identify the donor atoms, ligand denticity, coordination number, and expected coordination geometry around the cobalt center.
5. Explain how Co(salen) binds oxygen reversibly. How is this process similar to and different from oxygen binding in hemoglobin?
6. During the oxygenation experiment, the product changes upon treatment with CH_2Cl_2 . Propose a chemical explanation for your observations.

Notes

Further Readings

1. Woollins, J. D., Ed. *Inorganic Experiments*, 3rd Revised Edition; John Wiley & Sons: Weinheim, Germany, 2010; pp. 167–179.
2. Davies, S. G. Preparation of *N,N'*-Ethylenebis(salicylideneiminato)cobalt(II) [Co(salen)]. *Inorganic Syntheses* **1970**, 12, 199–201.
3. Inorganic Chemistry Miessler, G. L.; Fischer, P. J.; Tarr, D. A. *Inorganic Chemistry*, 5th or 6th ed.; Pearson.

Techniques: Schiff base synthesis, inert-atmosphere synthesis, reflux, suction filtration, ATR-IR spectroscopy, ^1H NMR spectroscopy, and magnetic susceptibility measurements.

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4.9: Synthesis of Some Representative Metal Organic Frameworks (MOFs)

Learning Objectives

- Synthesize the metal–organic frameworks (MOFs) MOF-5, IRMOF-3, and HKUST-1 using solution-based or solvothermal methods.
- Explain the structural features of MOFs, including metal nodes, organic linkers, and framework topology.
- Characterize synthesized MOFs using powder X-ray diffraction (PXRD) and compare experimental patterns with reference data to assess crystallinity and phase purity.
- Use thermogravimetric analysis (TGA) to evaluate the thermal stability of MOFs and identify solvent loss and framework decomposition.
- Examine MOF crystals using optical microscopy and relate crystal morphology to synthesis conditions.

Introduction

Metal–organic frameworks (MOFs) have become an important and rapidly growing area of research in chemistry and materials science. This subclass of coordination polymers represents an interesting class of compounds with applications in fields such as gas storage and separation, catalysis, chemical sensing, drug delivery, and energy storage.

In this experiment, students will prepare three classic MOFs: the zinc-based frameworks MOF-5 and IRMOF-3, and the copper-based framework HKUST-1.

Experimental Method

Safety Precautions

- Wear approved safety goggles, a laboratory coat, and appropriate chemical-resistant gloves at all times.
- Perform all procedures involving DMF, dichloromethane (CH_2Cl_2), triethylamine, and ethanol in a properly functioning fume hood. Avoid skin contact and inhalation of vapors.
- Triethylamine and ethanol are highly flammable. Keep them away from ignition sources and use only hot plates or heating mantles for heating.
- Inspect all glassware before use, and securely clamp reflux apparatus before heating. Use caution when handling hot glassware and samples removed from the oven.
- Dispose of all solvent and metal-containing waste in the designated waste containers. Do not pour chemicals or solvents down the drain.

Materials

Chemical Name	Formula	CAS Number	Molar Mass (g/mol)	Density (g/mL)
Zinc acetate dihydrate	$\text{Zn}(\text{OAc})_2 \cdot 2\text{H}_2\text{O}$	5970-45-6	219.50	— (solid)
Zinc nitrate hexahydrate	$\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	10196-18-6	297.49	— (solid)
Copper(II) nitrate trihydrate	$\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	10031-43-3	241.60	— (solid)
Terephthalic acid	$\text{C}_8\text{H}_6\text{O}_4$	100-21-0	166.13	— (solid)
1,3,5-Benzenetricarboxylic acid (Trimesic acid)	$\text{C}_9\text{H}_6\text{O}_6$	554-95-0	210.14	— (solid)
2-Amino-1,4-benzenedicarboxylic acid	$\text{C}_8\text{H}_7\text{NO}_4$	610-05-1	181.15	— (solid)
<i>N,N</i> -Dimethylformamide (DMF)	$\text{C}_3\text{H}_7\text{NO}$	68-12-2	73.09	0.944

Triethylamine	C ₆ H ₁₅ N	121-44-8	101.19	0.726
Ethanol	C ₂ H ₆ O	64-17-5	46.07	0.789
Dichloromethane	CH ₂ Cl ₂	75-09-2	84.93	1.325
Deionized water	H ₂ O	7732-18-5	18.02	1.000

Procedure

MOF-5

Day 1

Add terephthalic acid (0.200 g), DMF (16 mL), and triethylamine (0.34 mL) to a 50 mL round-bottom flask. In a separate container, prepare a solution of zinc acetate dihydrate (Zn(OAc)₂·2H₂O) (0.679 g) in DMF (20 mL). Add the zinc solution dropwise to the organic solution over 5 min with stirring. Continue stirring the reaction mixture for 90 min. Filter the precipitate using Büchner filtration. Transfer the solid to a vial and cover it with 10 mL of DMF.

Day 2

Carefully remove the DMF from the sample using a pipette. Replace it with an equal volume of fresh DMF. Do not allow the sample to dry during the solvent exchange, as drying may reduce the crystallinity and accessible porosity of the MOF.

HKUST-1

Day 1

Add copper(II) nitrate (1.75 g), 1,3,5-benzenetricarboxylic acid (0.84 g), and ethanol (50 mL) to a 100 mL round-bottom flask. Heat the mixture under reflux with stirring for 48 h.

Day 2

Turn off the heat and stirring, and allow the reaction mixture to cool to room temperature. Filter the precipitate using Büchner filtration and wash it with deionized water (2 × 20 mL), followed by ethanol.

IRMOF-3

Day 1

Add zinc nitrate hexahydrate (0.320 g), 2-amino-1,4-benzenedicarboxylic acid (0.076 g), and *N,N*-dimethylformamide (DMF, 10 mL) to a 20 mL scintillation vial. Cap the vial and give it to the TA. The sample will be heated in an oven at 100 °C for 18 h.

Day 2

Carefully remove the mother liquor from the vial using a pipette. Wash the product with fresh DMF (10 mL), then replace the DMF with dichloromethane (CH₂Cl₂, 10 mL). Do not allow the sample to dry during the solvent exchange, as drying may reduce the crystallinity and accessible porosity of the MOF.

Data Collection and Analysis

1. Describe the color of the products.
2. Calculate the % yield. The calculation must be in your laboratory notebook.
3. Characterize the synthesized MOFs using optical microscopy, PXRD, and TGA according to the instructions provided by your TA.

The drying conditions used for MOF samples depend on the characterization technique.

Optical Microscopy

- No special sample preparation is required.

Powder X-ray Diffraction (PXRD)

- Remove excess solvent by decanting or using a pipette.
- Allow the sample to air-dry briefly or dry gently under a stream of nitrogen until no free solvent is visible.

- Do not over-dry MOF-5 or IRMOF-3 prior to PXRD, as excessive drying may reduce the crystallinity of these materials.

Thermogravimetric Analysis (TGA)

- **MOF-5 and IRMOF-3:** After solvent exchange, dry the sample under vacuum at room temperature or at 40–60 °C until the solid is free-flowing. Avoid prolonged heating or temperatures above 60 °C before analysis.
- **HKUST-1:** After washing with ethanol, dry the sample *in vacuo* at 80–100 °C for 6–12 h before analysis.

Questions and Further Thoughts

1. Using the primary literature or another reliable scientific source, locate the crystal structures of MOF-5, HKUST-1, and IRMOF-3. Draw the framework structure of each MOF, identifying the metal node (secondary building unit) and the organic linker.
2. Explain the role of triethylamine in the synthesis of MOF-5.
3. Compare the coordination environments of the metal ions in MOF-5, HKUST-1, and IRMOF-3.
4. Compare your experimental PXRD patterns with the reference patterns. What do they reveal about the crystallinity and phase purity of each MOF?
5. Identify the major mass-loss events in the TGA traces and explain what process each represents.
6. Compare the crystal size and morphology of the three MOFs using your optical microscopy observations. What factors may account for the differences you observed?
7. Why are MOFs often described as porous materials? Explain how their structures give rise to porosity and discuss why this property is important for applications such as gas storage, separations, or catalysis.

Notes

Further Readings

1. Sumida, K.; Arnold, J. Preparation, Characterization, and Postsynthetic Modification of Metal–Organic Frameworks: Synthetic Experiments for an Undergraduate Laboratory Course in Inorganic Chemistry. *J. Chem. Educ.* **2011**, *88* (1), 92–94. <https://doi.org/10.1021/ed100001s>.
2. Yang, D.; Gates, B. C. Catalysis by Metal Organic Frameworks: Perspective and Suggestions for Future Research. *ACS Catal.* **2019**, *9* (3), 1779–1798. <https://doi.org/10.1021/acscatal.8b04515>.
3. Panda, D.; Patra, S.; Awasthi, M. K.; Singh, S. K. Lab Cooked MOF for CO₂ Capture: A Sustainable Solution to Waste Management. *J. Chem. Educ.* **2020**, *97* (4), 1101–1108. <https://doi.org/10.1021/acs.jchemed.9b00337>.
4. Sakamaki, Y.; Tsuji, M.; Heidrick, Z.; Watson, O.; Durchman, J.; Salmon, C.; Burgin, S. R.; Beyzavi, H. Preparation and Applications of Metal–Organic Frameworks (MOFs): A Laboratory Activity and Demonstration for High School and/or Undergraduate Students. *J. Chem. Educ.* **2020**, *97* (4), 1109–1116. <https://doi.org/10.1021/acs.jchemed.9b01166>.
5. Yuan, M.; Sun, C.-Y.; Liu, Y.-W.; Lu, Y.; Zhang, Z.; Li, X.-H.; Tian, H.-R.; Zhang, S.; Liu, S.-X. Synthesis, Characterization, and Property Investigation of a Metal–Organic Framework Encapsulated Polyoxometalate Guests: An Advanced Inorganic Chemistry Experiment. *J. Chem. Educ.* **2020**, *97* (11), 4152–4157. <https://doi.org/10.1021/acs.jchemed.0c00905>.

Techniques: Solvothermal synthesis, crystallization, filtration, optical microscopy, powder X-ray diffraction (PXRD), thermogravimetric analysis (TGA).

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